



Magnetic reconnection experiments at the National Ignition Facility: from fluids to kinetics?

Vicente Valenzuela-Villaseca

Department of Astrophysical Sciences, Princeton University, USA

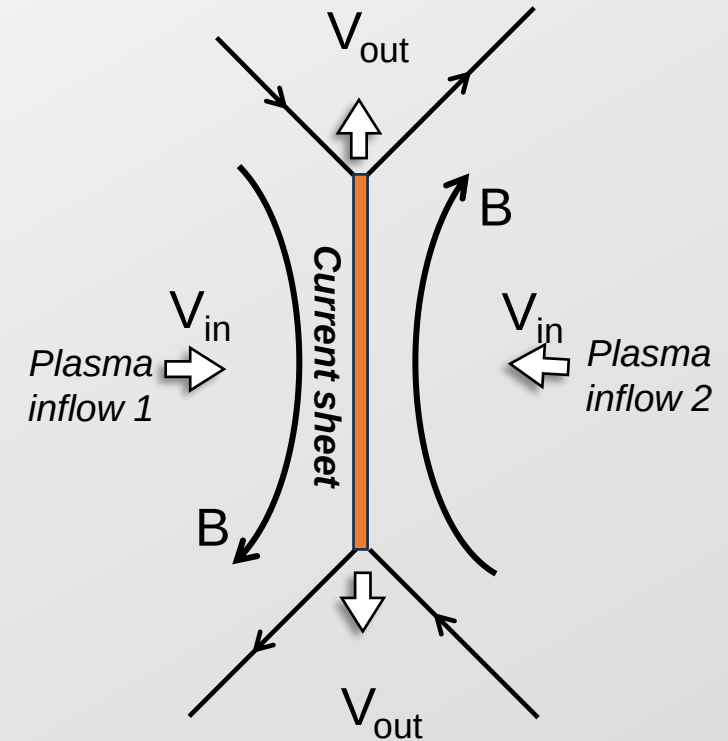
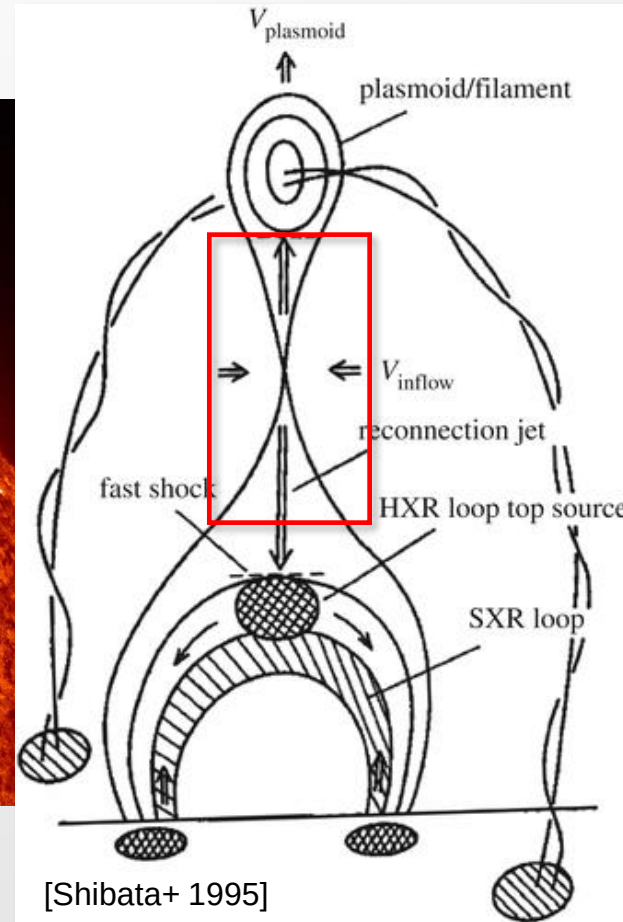
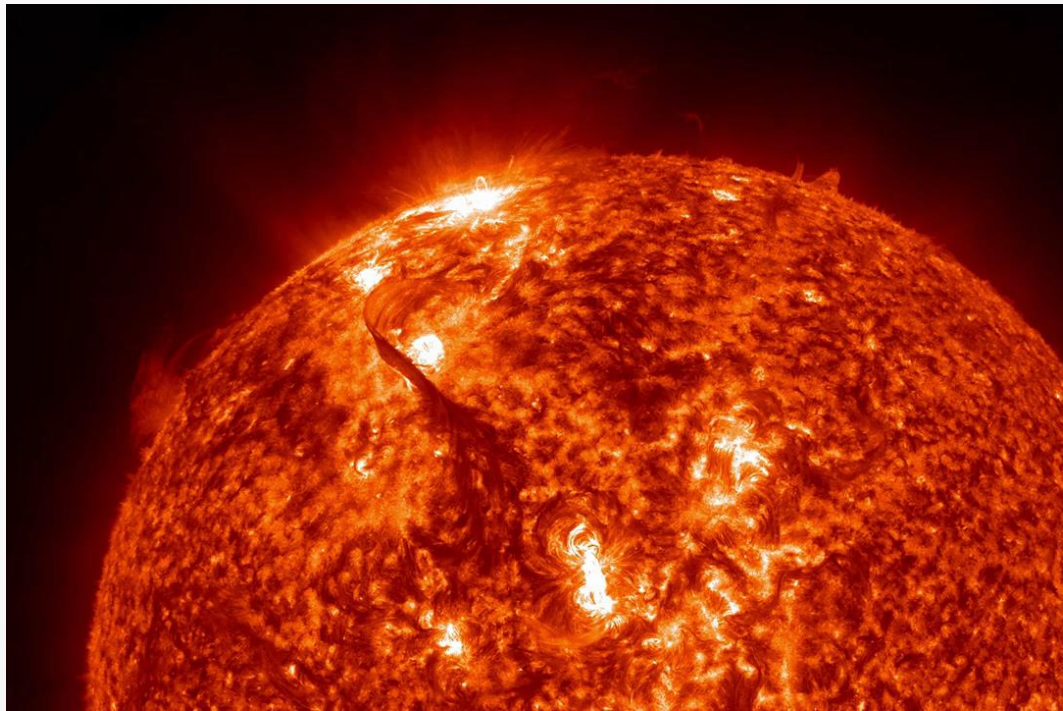


Magnetic fields: W. Fox+ *Phys. Rev. E* (accepted); [arxiv:2309.16165](https://arxiv.org/abs/2309.16165)
More magnetic fields: W. Fox+ *Phys. Rev. Lett.* (submitted); [arxiv:2003.06351](https://arxiv.org/abs/2003.06351)
Electron heating: V. Valenzuela-Villaseca+ *Phys. Plasmas* (submitted); [arxiv:2404.07799](https://arxiv.org/abs/2404.07799)

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Magnetic reconnection explosively converts magnetic energy into plasma kinetic energy

Magnetic reconnection driven explosion on the surface of the Sun



The Sweet-Parker model predicts that the reconnection mediated by resistivity and its rate is determined by the Lundquist number

Reconnection rate:
$$\frac{\text{\# of lines out}}{\text{\# of lines in}} = \frac{V_{\text{out}}}{V_{\text{in}}}$$

Conservation of mass:

$$\rho V_{\text{in}} L = \rho V_{\text{out}} \delta \rightarrow \frac{L}{\delta} = \frac{V_A}{V_{\text{in}}}$$

Conservation of energy:
(max outwards speed)

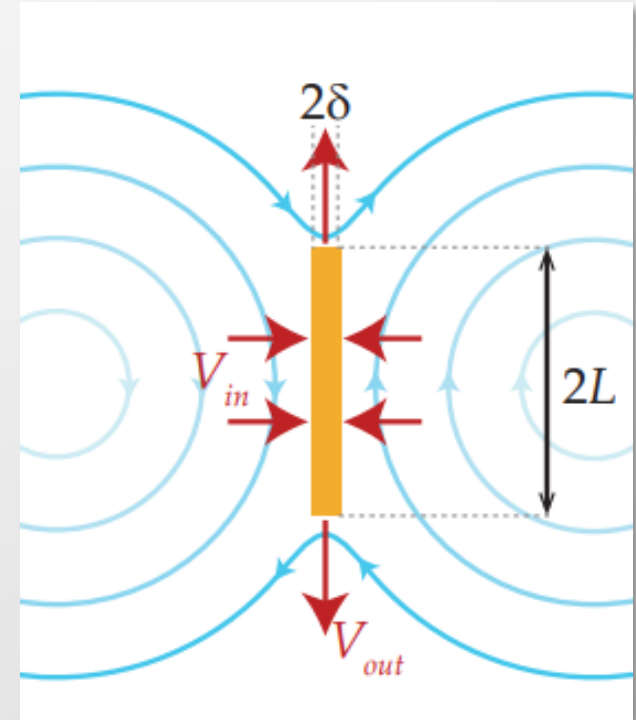
$$\frac{B^2}{2\mu_0} = \frac{1}{2} \rho V_{\text{out}}^2 \rightarrow V_{\text{out}} = \frac{B}{\sqrt{\mu_0 \rho}} \equiv V_A$$

Ampere's law: $J_z \sim B/\mu_0 \delta$

Ohm's law

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = \eta \mathbf{J} \rightarrow V_{\text{in}} B = \eta J_z$$

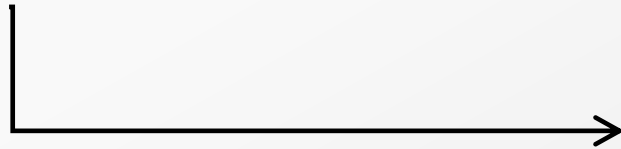
$$\rightarrow \frac{V_{\text{out}}}{V_{\text{in}}} = \left(\frac{\mu_0 L V_A}{\eta} \right)^{1/2} \equiv S^{1/2}$$



1. S is called **Lundquist number**, it is dimensionless (universal), and **sets the reconnection rate**
2. SP magnetic reconnection **dissipates energy through Spitzer resistivity**

The Sweet-Parker picture has dominated the field, but fails at explaining the explosiveness of reconnection in nature

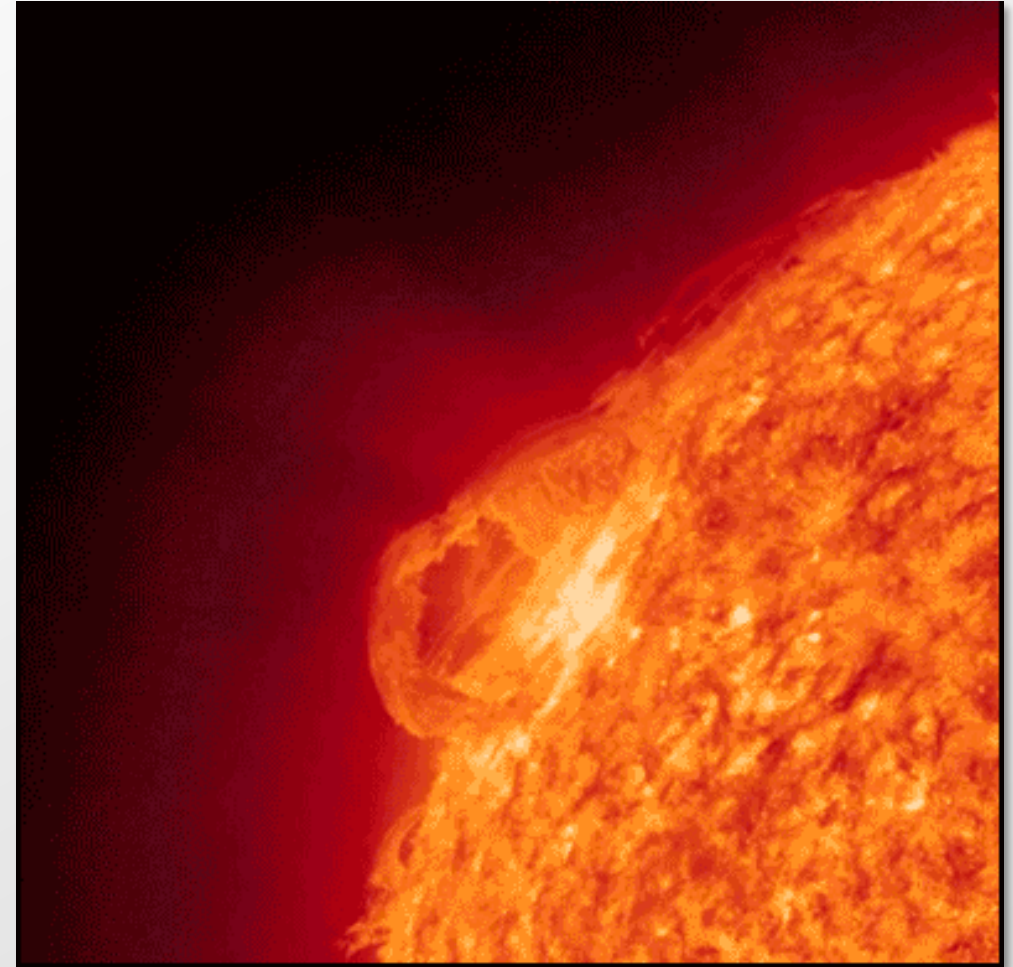
Prediction: 1000 yrs. **Reality: 10 minutes!**



Key questions in magnetic reconnection

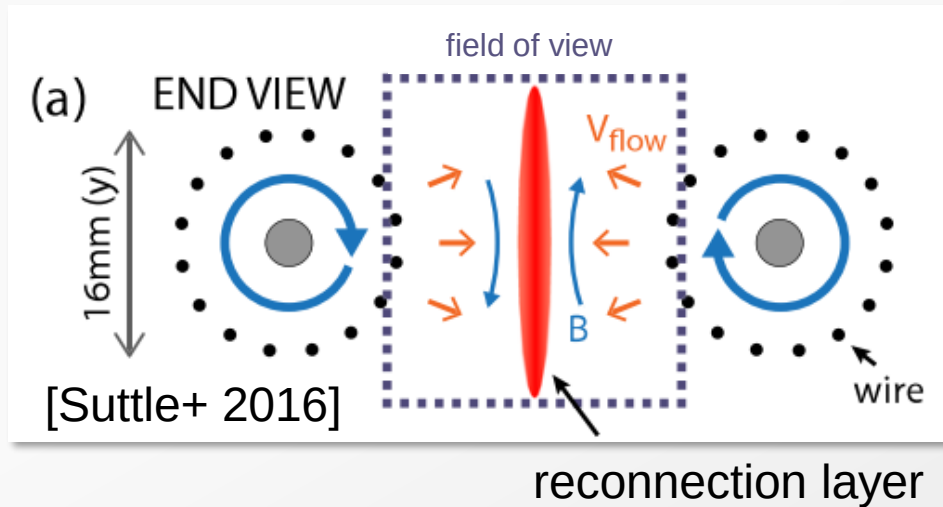
[Zweibel & Yamada 2016]

1. Why is magnetic reconnection fast/explosive/bursty?
2. Onset: What triggers reconnection? On what scales (kinetic, MHD)? Is it universal?



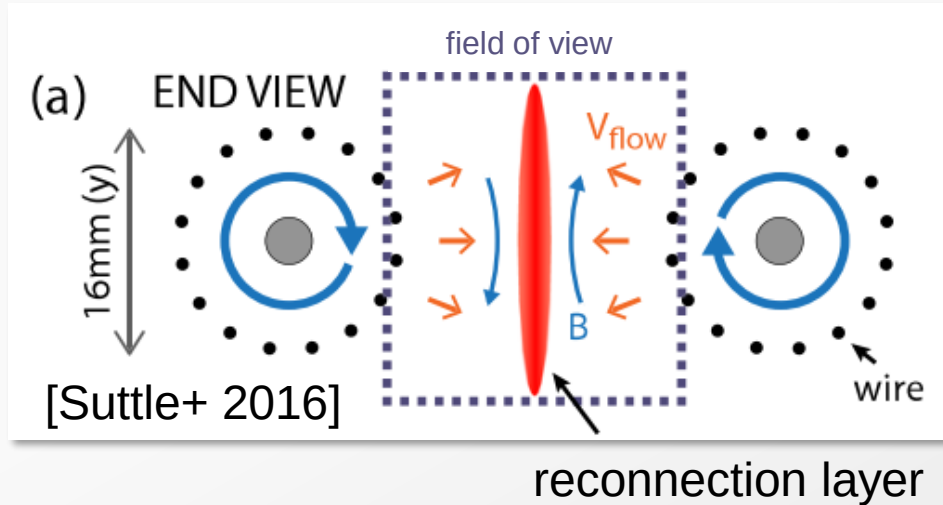
Magnetic reconnection is not a steady process, it can become subject to the tearing “plasmoid” instability

Pulsed-power magnetic
reconnection experiments

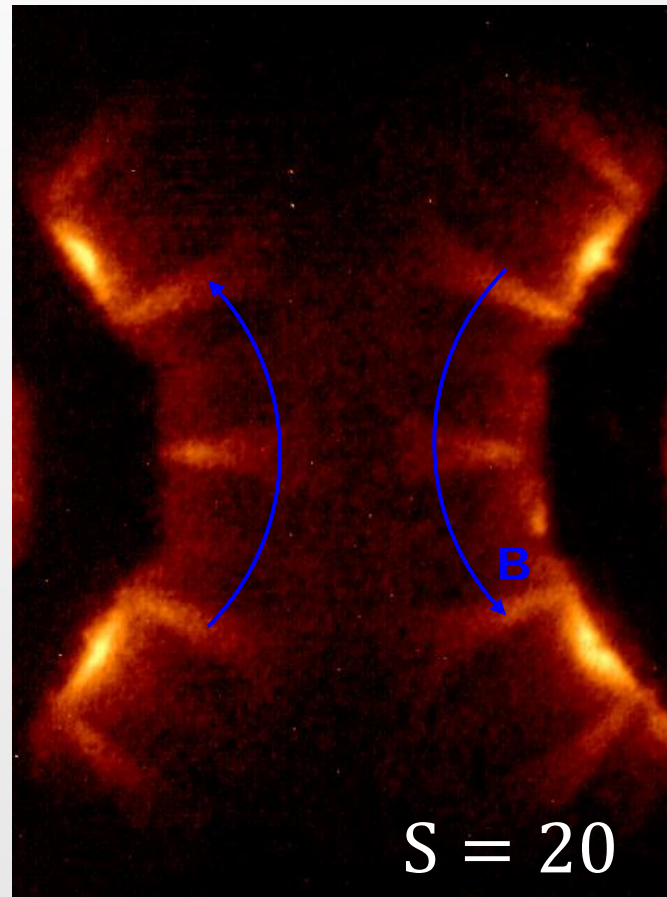


Magnetic reconnection is not a steady process, it can become subject to the tearing “plasmoid” instability

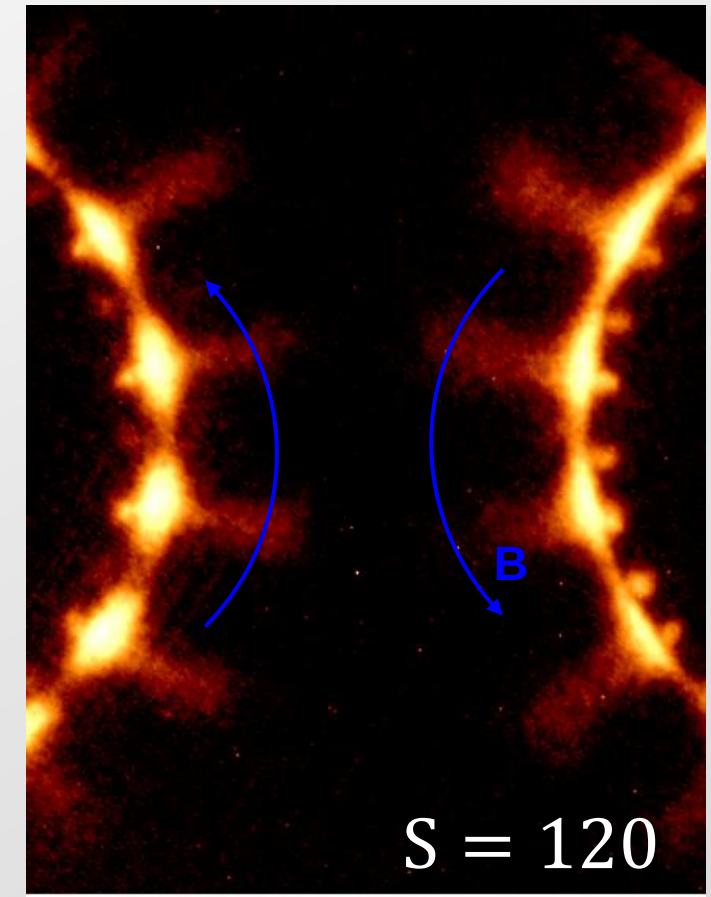
Pulsed-power magnetic reconnection experiments



Stable SP with aluminium plasma experiments
[Suttle+ 2016]

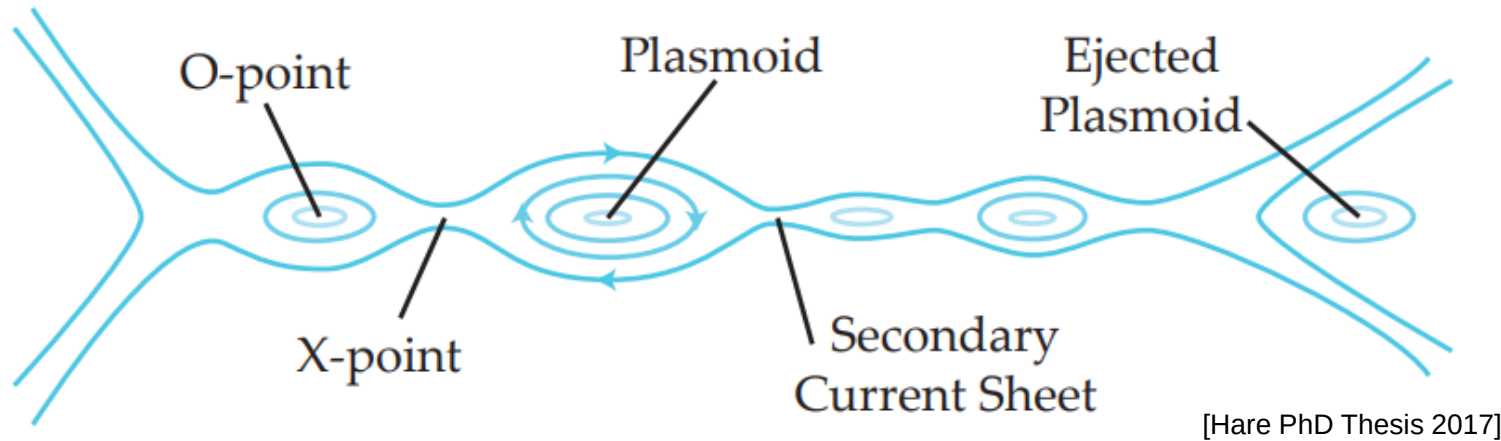


Plasmoid unstable layer in carbon plasma
[Hare+ 2017]



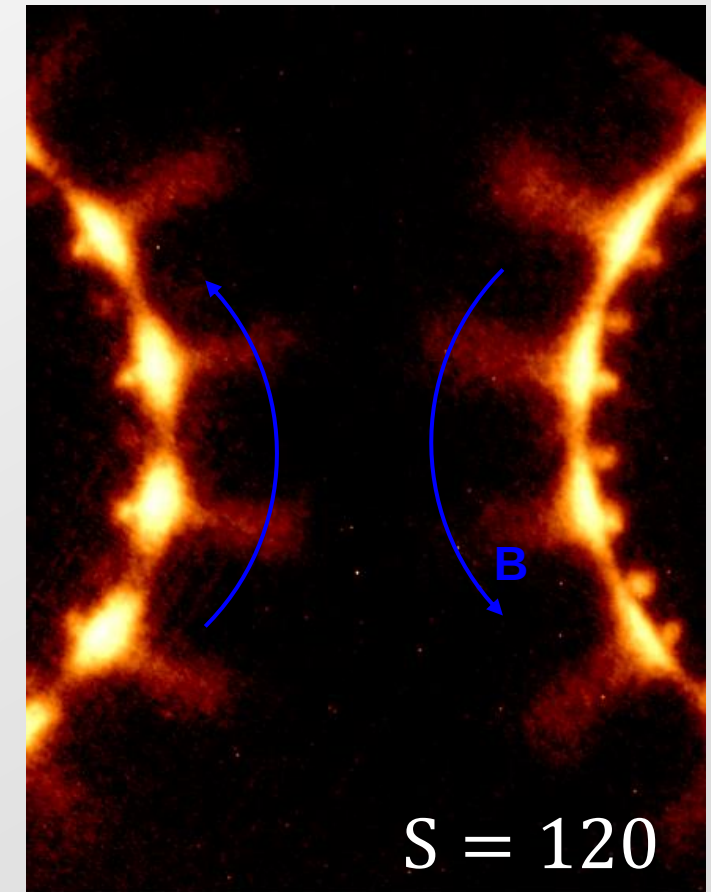
Magnetic reconnection is not a steady process, it can become subject to the tearing “plasmoid” instability

$$S \sim \frac{L \cdot B \cdot T_e^{3/2}}{\sqrt{n_e}} > S_c$$



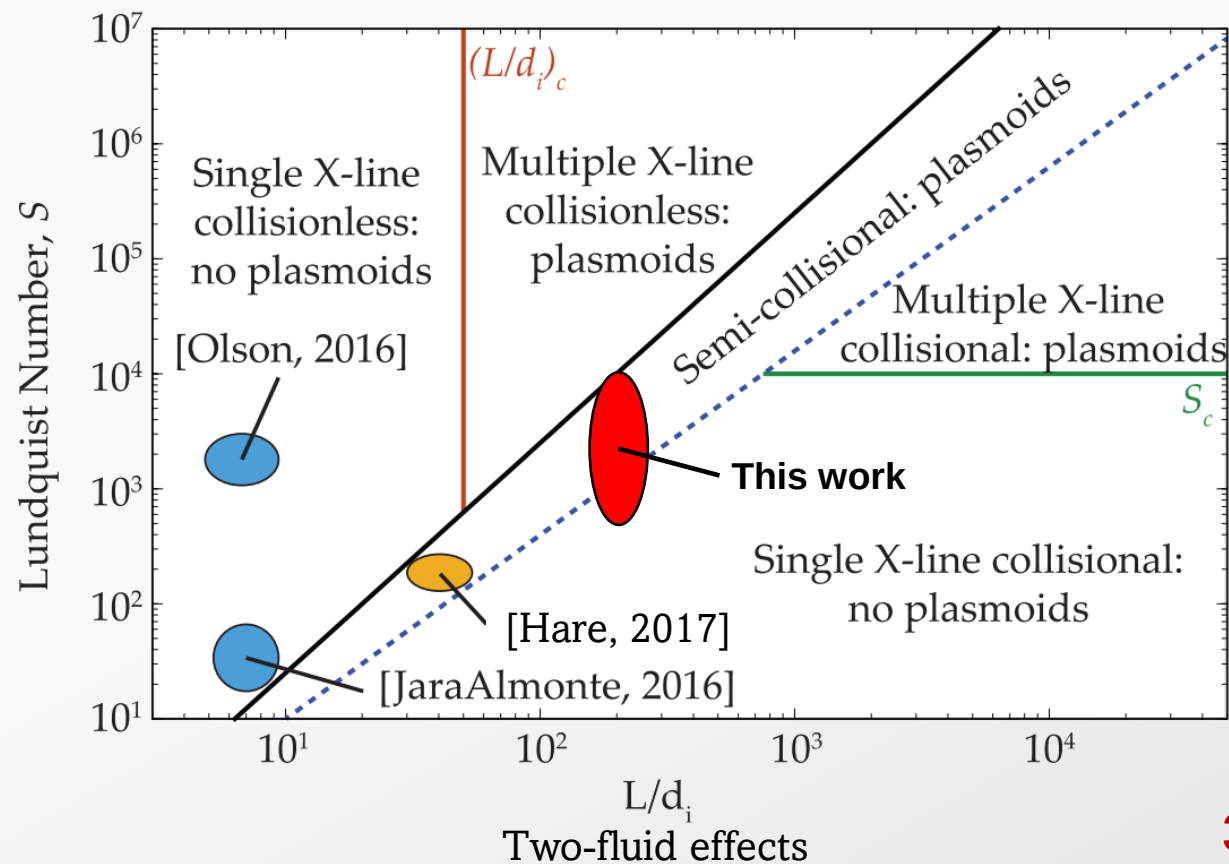
Current sheet length L is not a free parameter
in Sweet-Parker reconnection

Plasmoid unstable layer in
carbon plasma
[Hare+ 2017]



$S = 120$

Plasmoid-mediated reconnection can operate in a large parameter space



Adapted from [Hare PhD Thesis, 2017;
c.f. Loureiro & Uzdensky 2016]

Plasmoid reconnection depends on

- Particle magnetization
- Heating/cooling rates
- Collisionality (electron trapping, temperature anisotropy, etc)
- Nonlinear coupling: stochastic plasmoid chains (turbulence), monster plasmoids, etc.

Key questions in magnetic reconnection (continued)
[Zweibel & Yamada 2016]

3. What is the energy partition?

4. What is the interplay of multiscale processes?

An ecosystem of laboratory experiments allow accessing different regimes of magnetic reconnection

Magnetically Driven

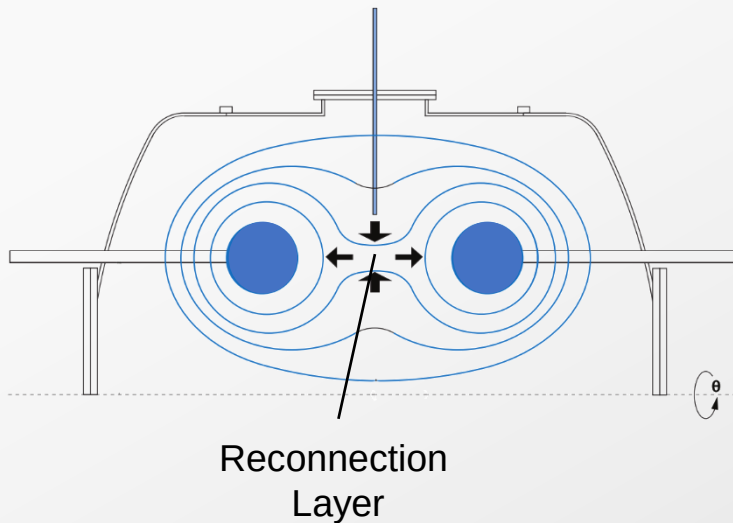
$$B=0.03 \text{ T}, n_e=5 \times 10^{13} \text{ cm}^{-3},$$

$$V=10 \text{ km/s}, L=10 \text{ cm}$$

$$T_e=10 \text{ eV}, T_i=10 \text{ eV}$$

$$\beta_{th} \ll 1, \beta_{dyn} \ll 1$$

Strongly magnetized



Pulsed Power Driven

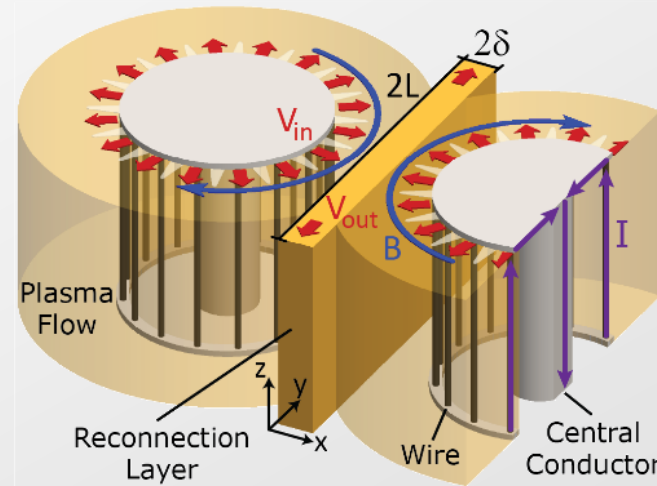
$$B=3 \text{ T}, n_e=5 \times 10^{17} \text{ cm}^{-3},$$

$$V=50 \text{ km/s}, L=1 \text{ cm}$$

$$T_e=100 \text{ eV}, T_i=500 \text{ eV},$$

$$\beta_{th} \sim 1, \beta_{dyn} \sim 1$$

Unity beta



Laser Driven

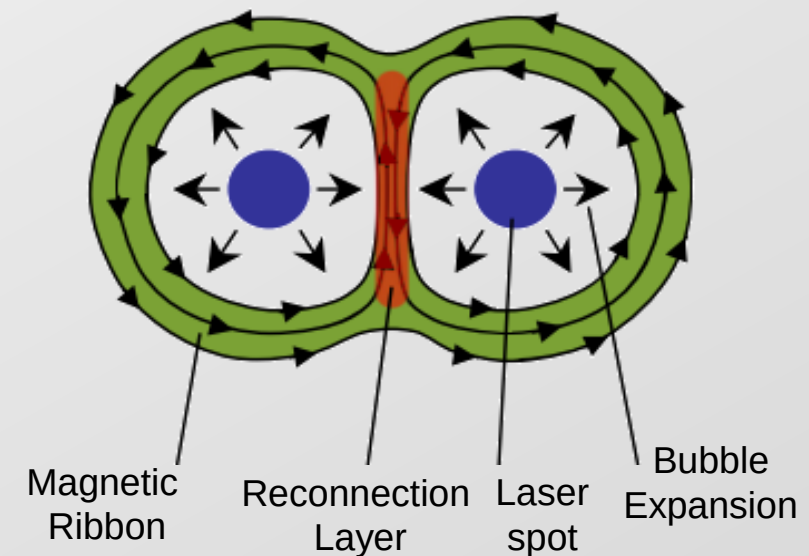
$$B=50 \text{ T}, n_e=7 \times 10^{19} \text{ cm}^{-3},$$

$$V=500 \text{ km/s}, L=0.1 \text{ cm},$$

$$T_e=650 \text{ eV}, T_i=250 \text{ eV}$$

$$\beta_{th} \gg 1, \beta_{dyn} \gg 1$$

Weakly magnetized



Laser-driven magnetic reconnection experiments employ the self-magnetization of plasma plumes

Extended Ohm's
law

$$m_e \mathbf{\dot{v}}_e = -\nabla p - en_e(\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) \approx 0$$

Faraday's law

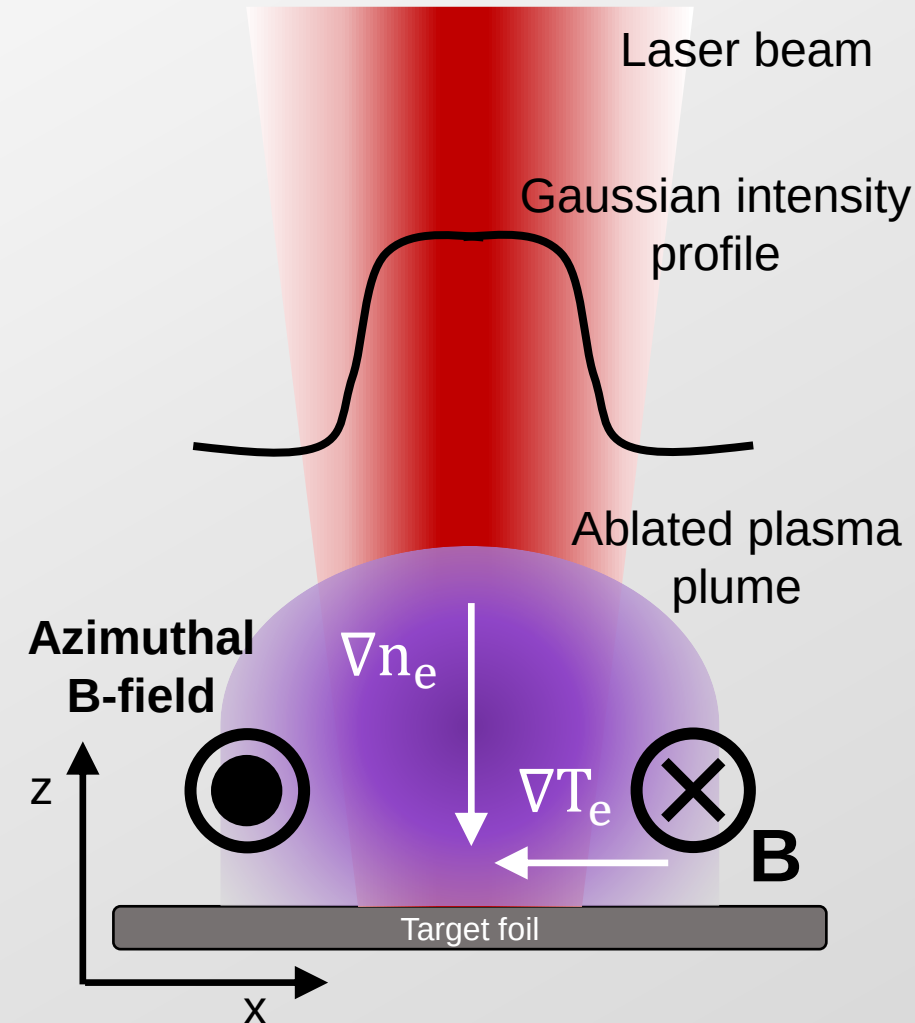
$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}$$

Extended induction
equation

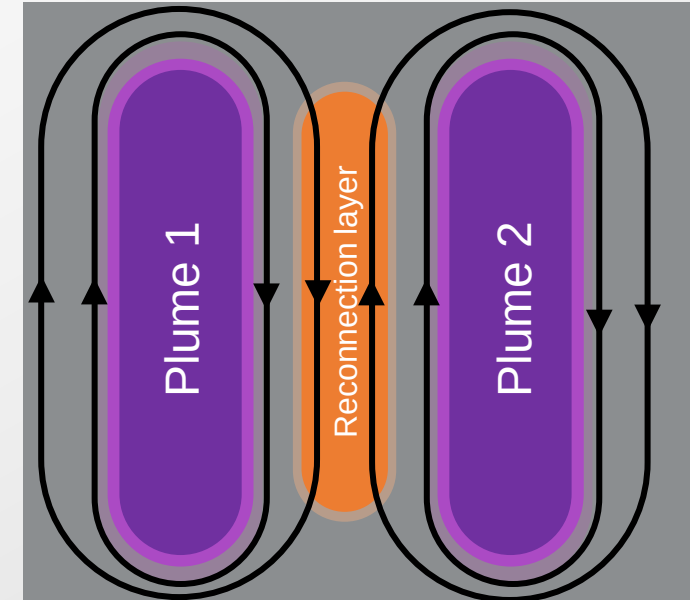
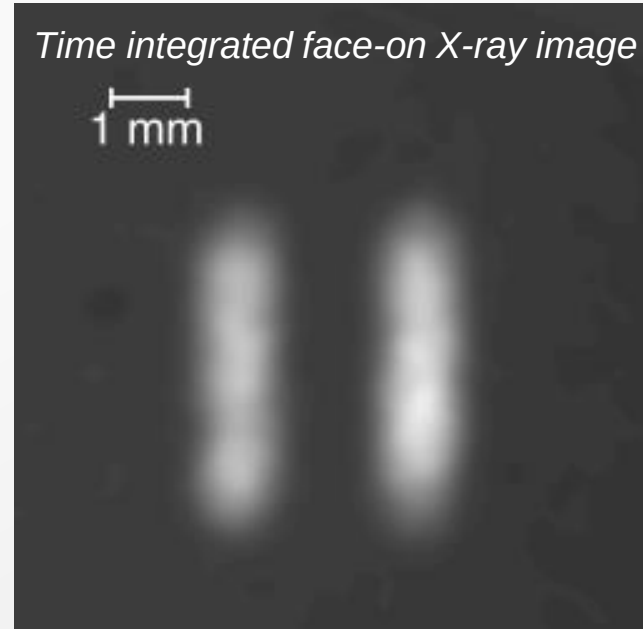
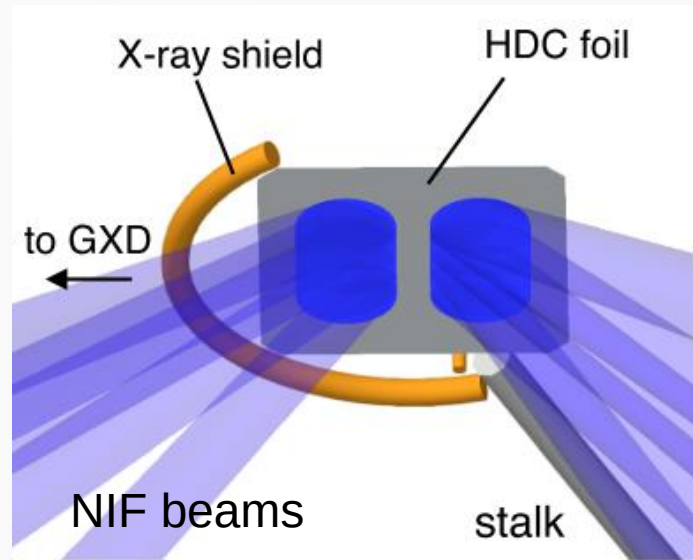
$$\frac{\partial \mathbf{B}}{\partial t} = \underbrace{\nabla \times \mathbf{u} \times \mathbf{B}}_{\text{Flux freeze}} + \underbrace{\nabla \times \frac{\mathbf{J} \times \mathbf{B}}{en_e}}_{\text{Hall effect}} + \underbrace{\frac{\nabla T_e \times \nabla n_e}{en_e}}_{\text{Biermann battery}}$$

Transport Source

The Biermann battery effect magnetizes laser plasmas without external sources



We tiled 40 beams to create two elongated plasma plumes in the high-Lundquist kinetic regime



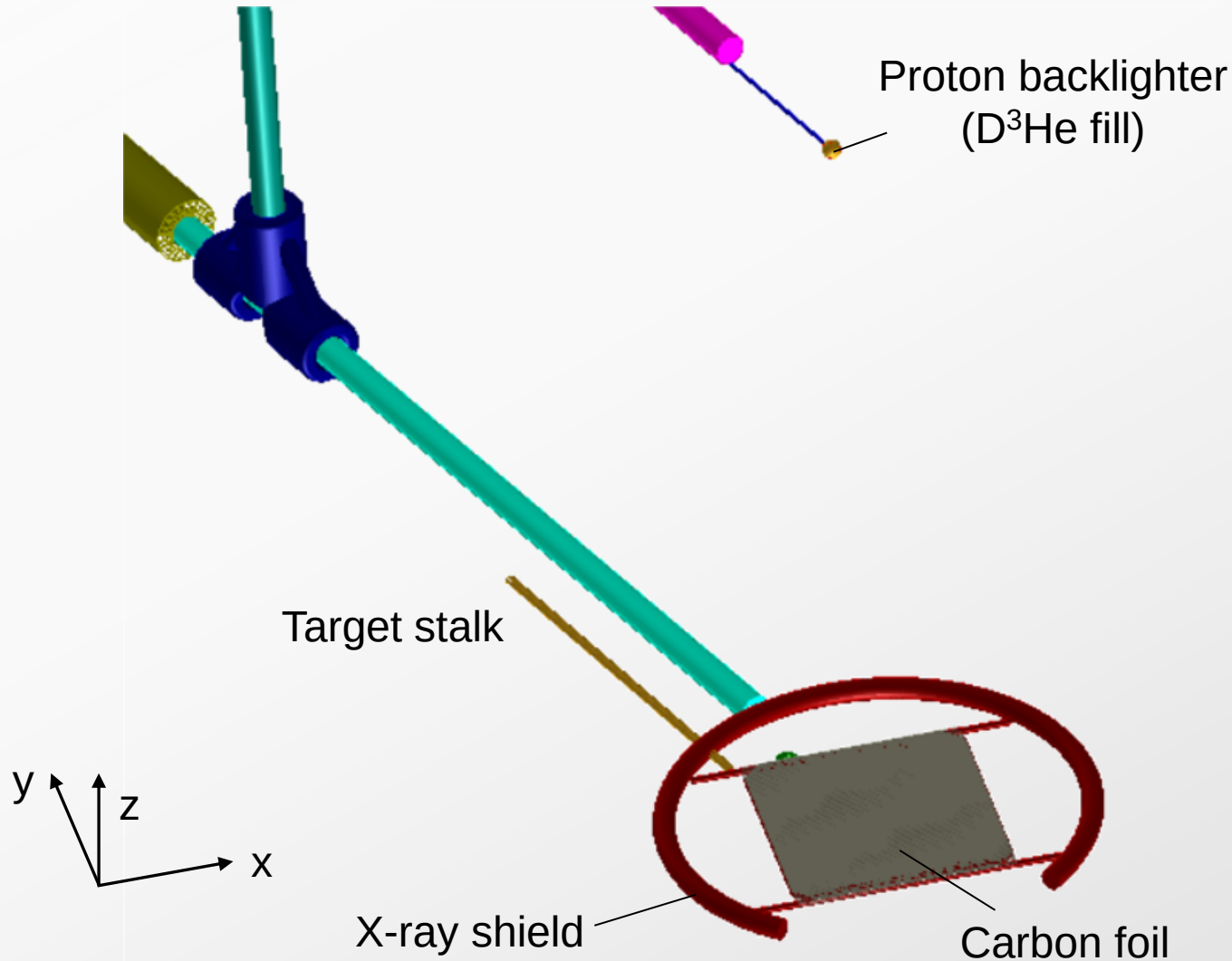
- 2.4 mm separation
- Total intensity: 10^{14} W/cm²
- 351 nm wavelength
- Laser duration: 0.9 ns

Typical plasma parameters:

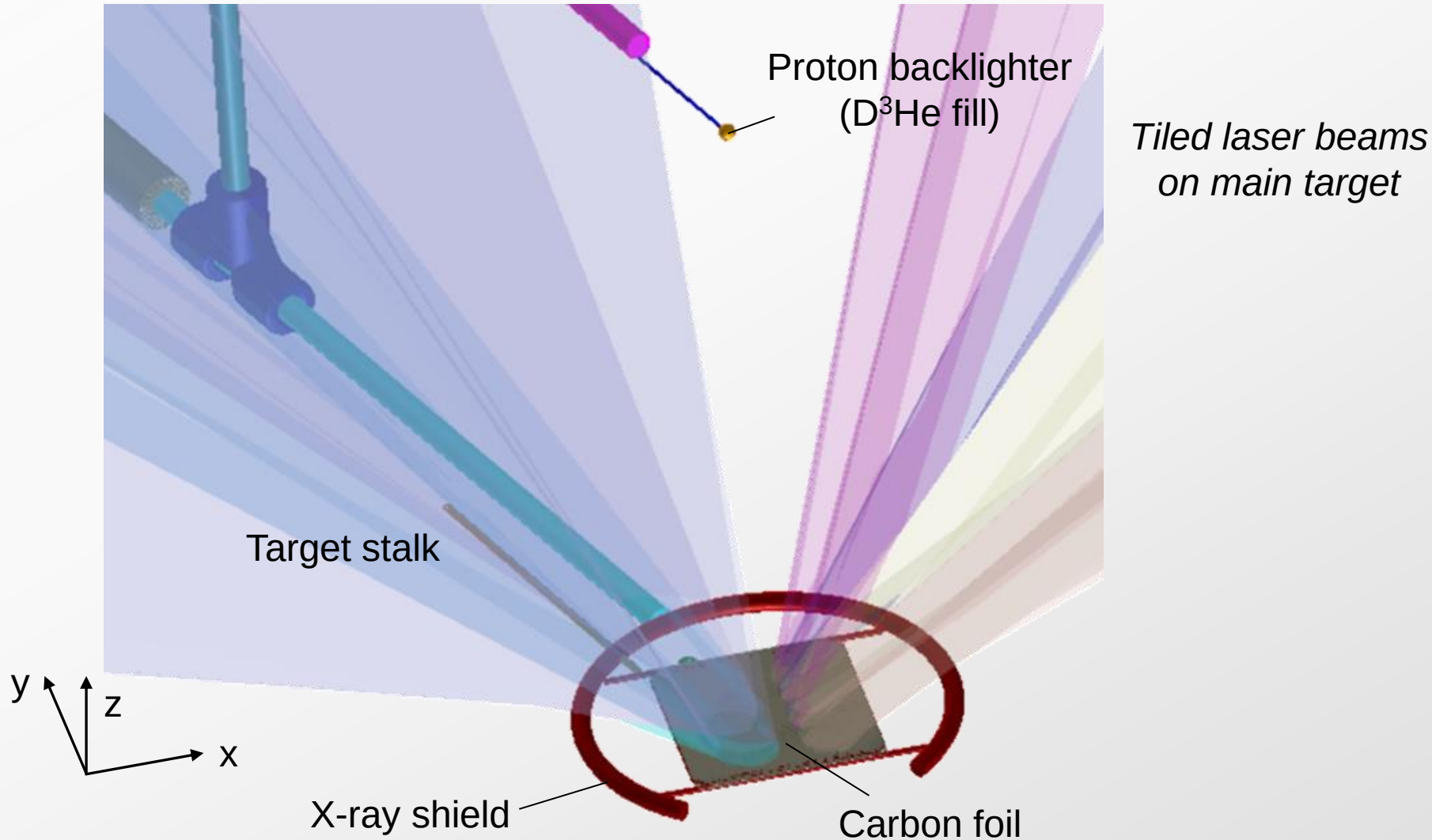
- $n_e \sim 1e19 - 1e20$ cm⁻³
- $T_e \sim 200 - 300$ eV
- $V_{in} \sim 230$ km/s
- $B \sim 5$ T

- Super-Alfvenic: $M_A = V_{in}/V_A \sim 5$
- Flow dominated:
 - $\beta_{th} = p_{th}/p_{mag} > 10$
 - $\beta_{dyn} = p_{ram}/p_{mag} > 50$
- High-Lundquist:
 - $S \sim 500 - 4000$

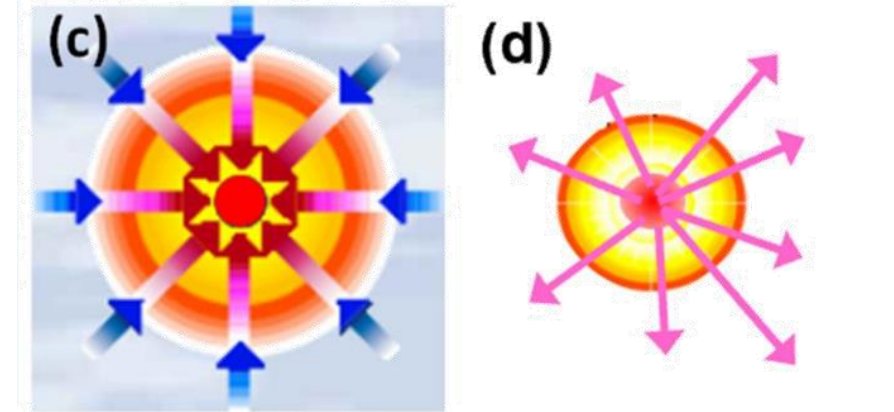
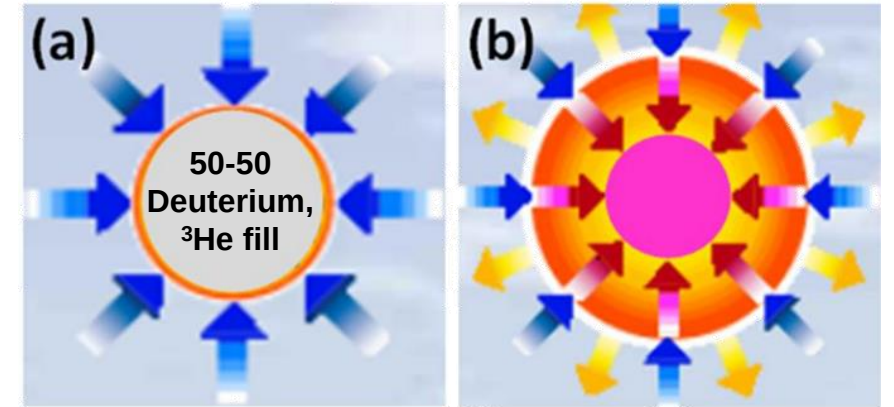
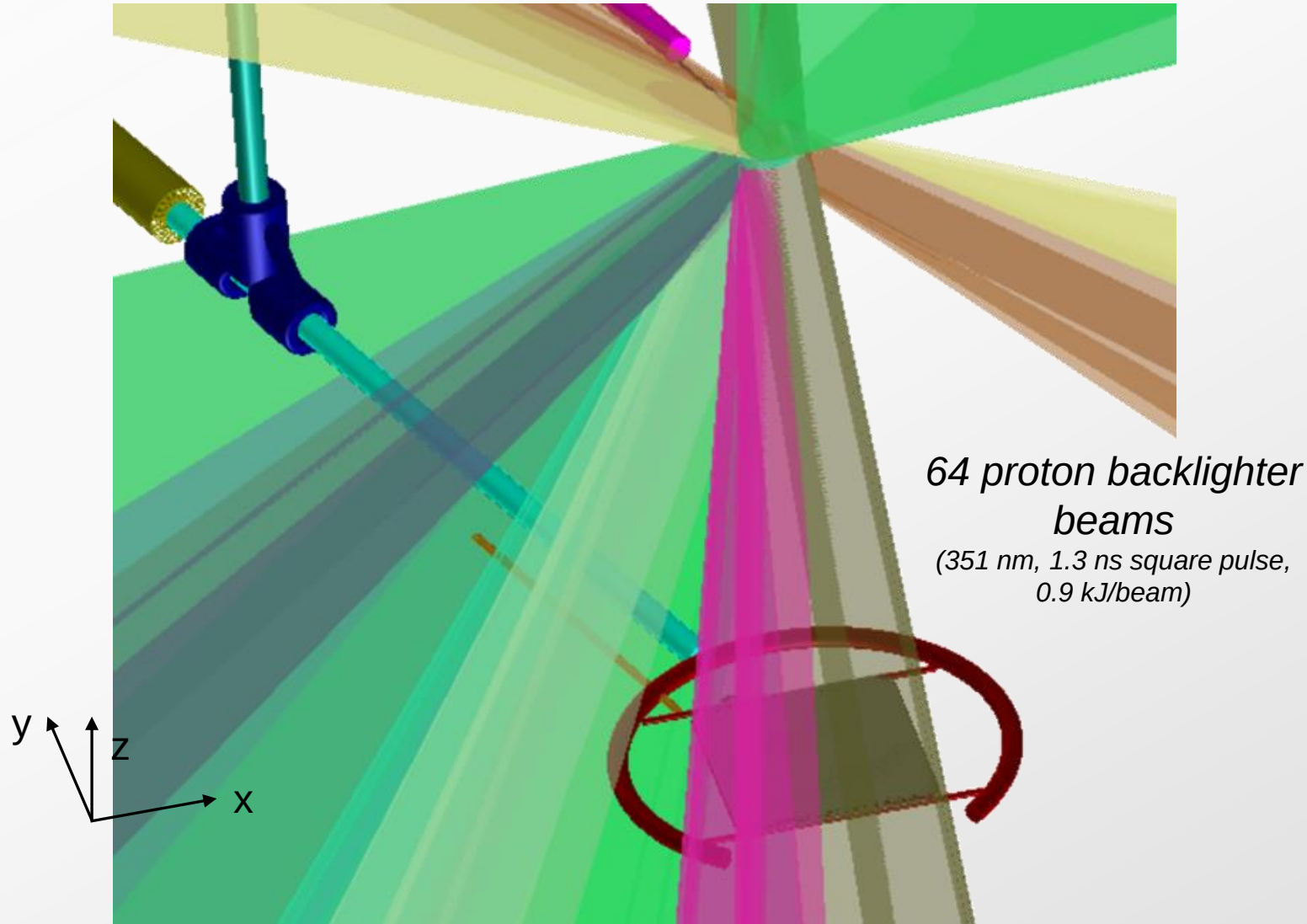
Magnetic field structure and evolution was probed using proton radiography from an ICF backlighter



Magnetic field structure and evolution was probed using proton radiography from an ICF backlighter

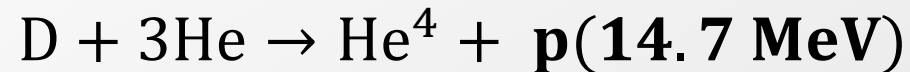
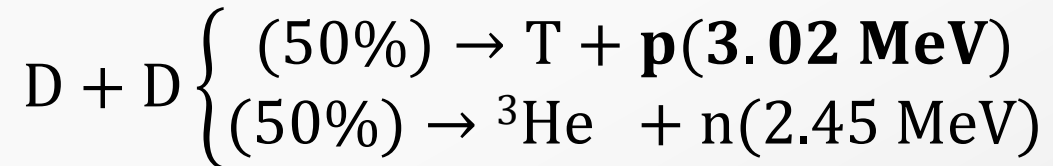


Magnetic field structure and evolution was probed using proton radiography from an ICF backlighter

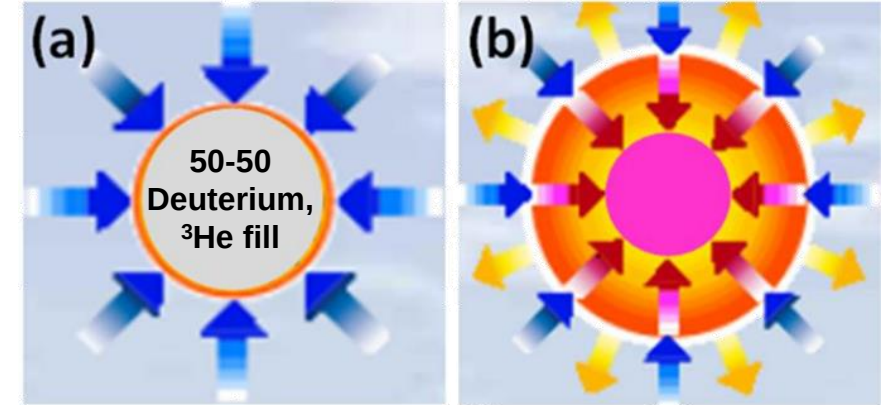


Magnetic field structure and evolution was probed using proton radiography from an ICF backlighter

Fusion reactions

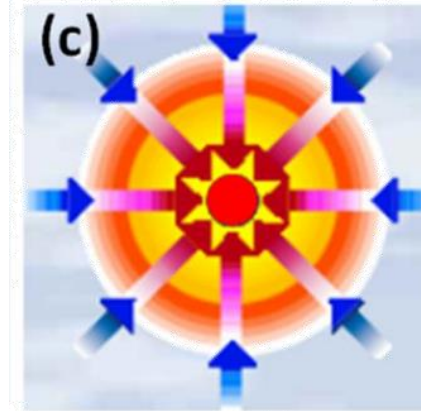


The proton beams can be used to probe electromagnetic fields in the plasma

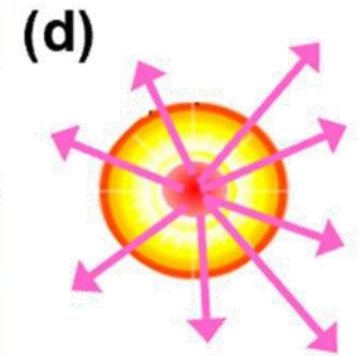


Spherical capsule is irradiated by laser beams

Outer shell explodes, forcing the capsule implosion

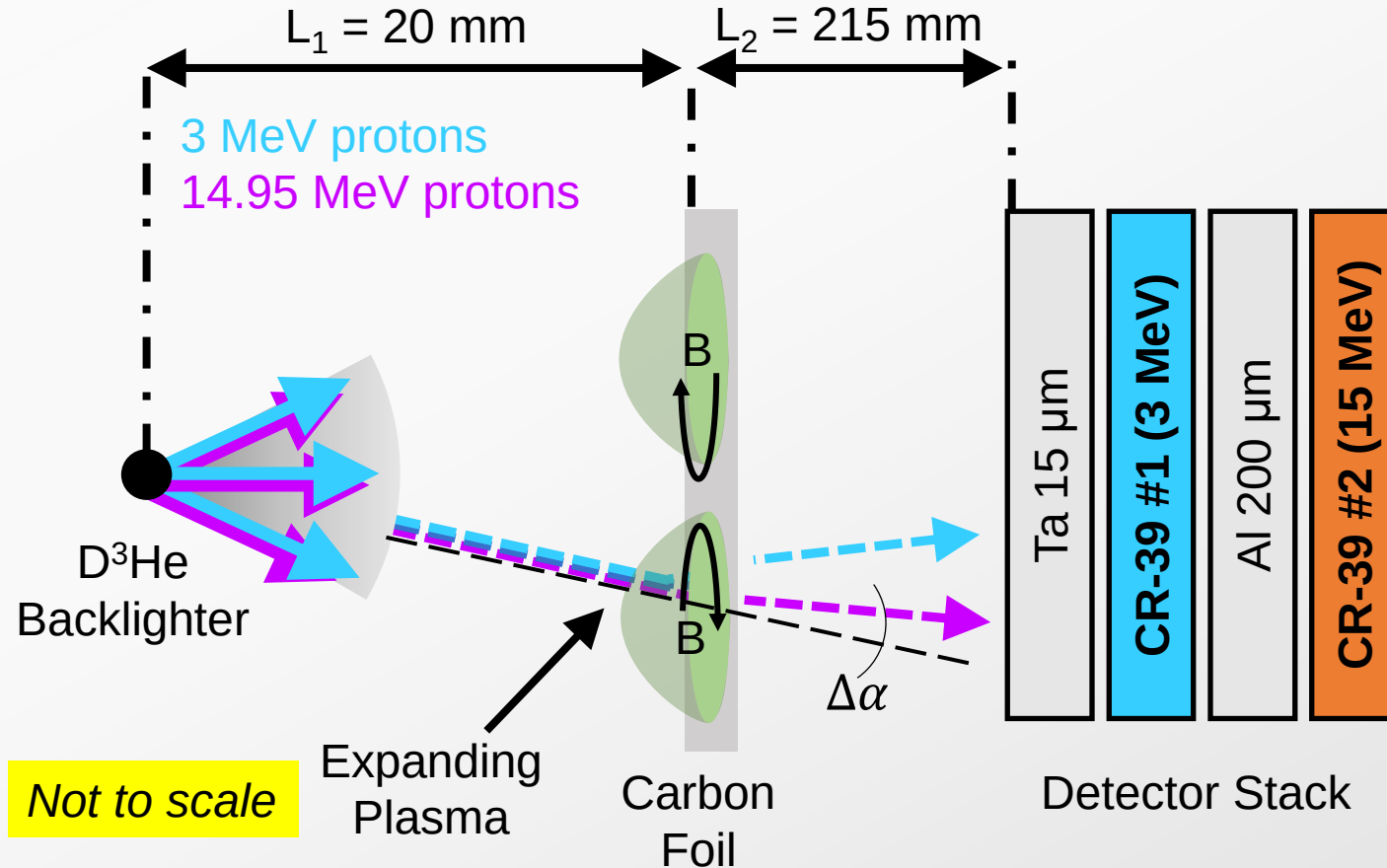


Compression at the centre creates a hotspot (40 μm diameter)



Fusion reactions provide monoenergetic proton beams

Magnetic field structure and evolution was probed using proton radiography from an ICF backlighter

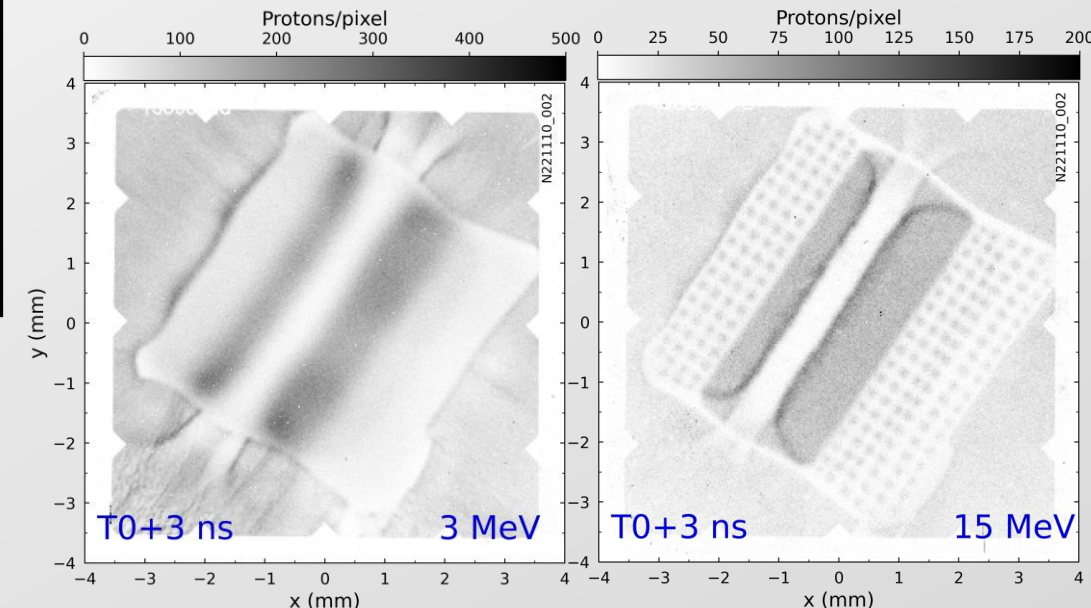


Adapted from [Griff-McMahon+ PRE (accepted)]

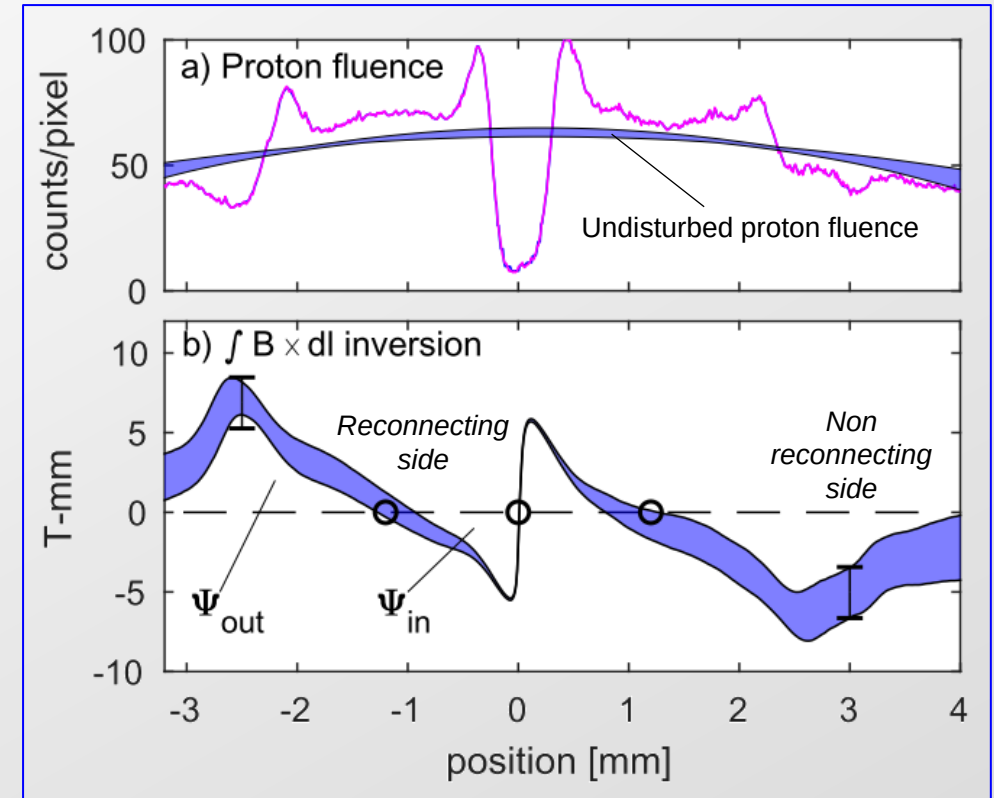
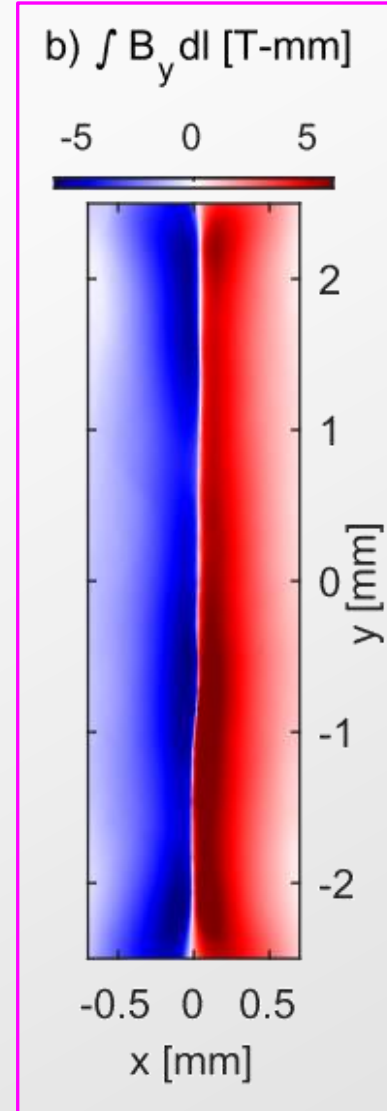
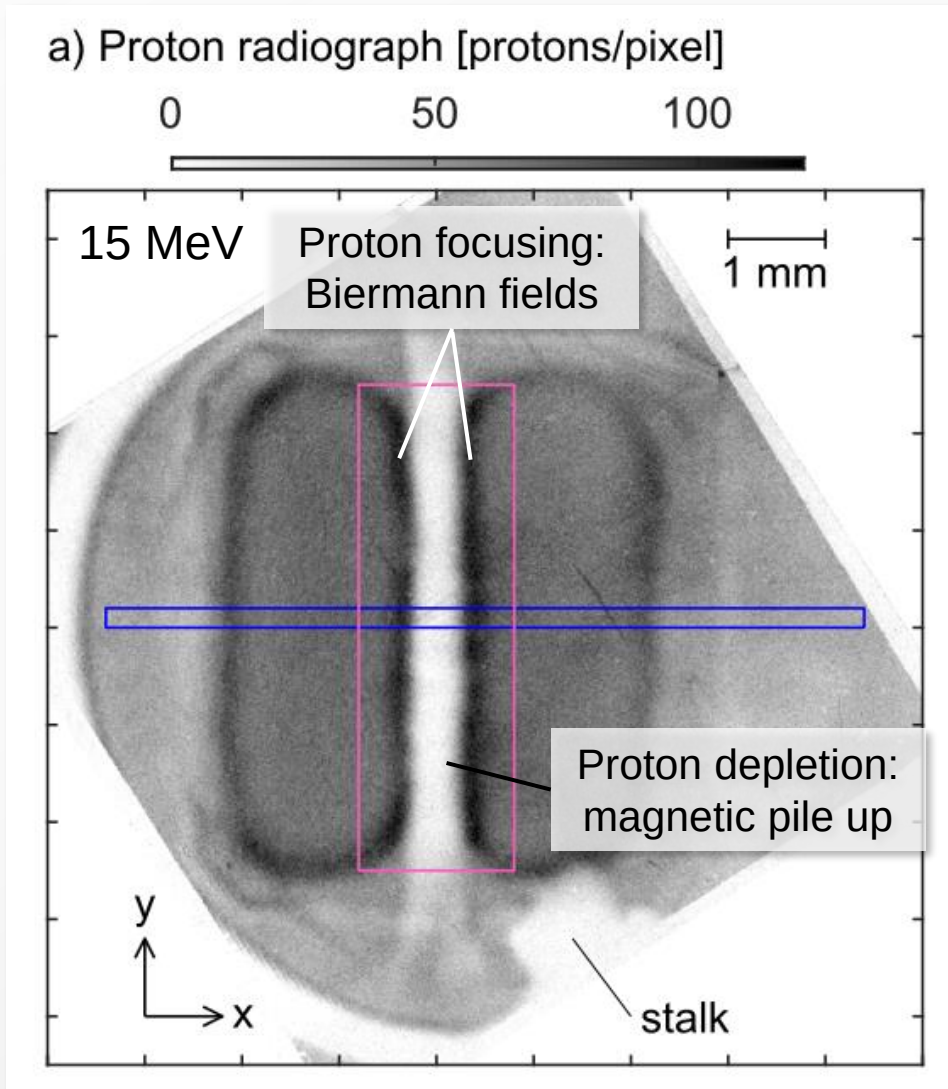
- Proton deflection due to E- and B-fields scales differently with proton energy.

$$\Delta\alpha = \frac{e}{2E_p} \int (\mathbf{E} + \mathbf{V}_p \times \mathbf{B})_{\perp} d\ell$$

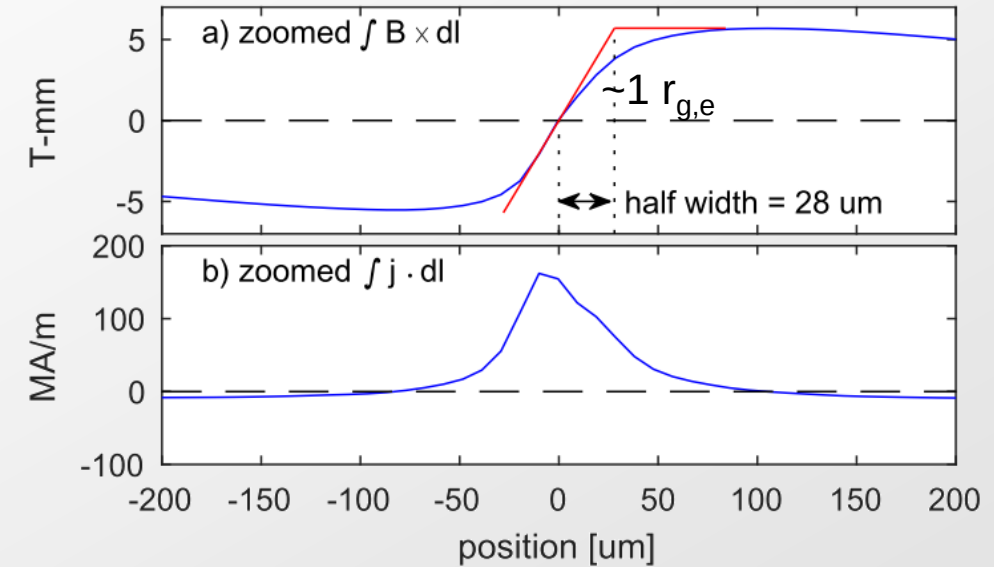
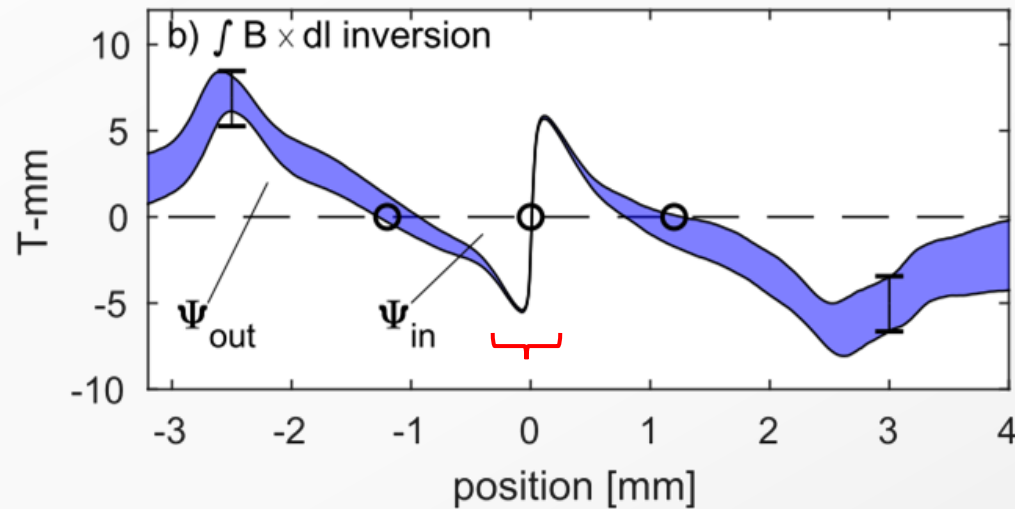
- Field reconstruction can disambiguate using the 2 monoenergetic beams



Proton radiography shows how the current sheet narrows to electron gyro-scales



The flux annihilation allows estimating the out-of-plane reconnected electric field and current density



Magnetic flux annihilation rate:

$$\Psi_{in} = 2.5 \pm 0.3 \text{ T} \cdot \text{mm}^2$$

$$\Psi_{out} = \int B_y dx d\ell = 5.5 - 15 \text{ T} \cdot \text{mm}^2$$

$$\rightarrow \Delta\Psi = \Psi_{out} - \Psi_{in} > 3 \text{ T} \cdot \text{mm}^2$$

Reconnected line-integrated electric field

$$\langle \int E_z d\ell \rangle = \Delta\Psi / \Delta t \sim 1300 \text{ V}$$

Reconnected line-integrated current density

$$\int j_z d\ell \sim 160 \text{ MA/m}$$

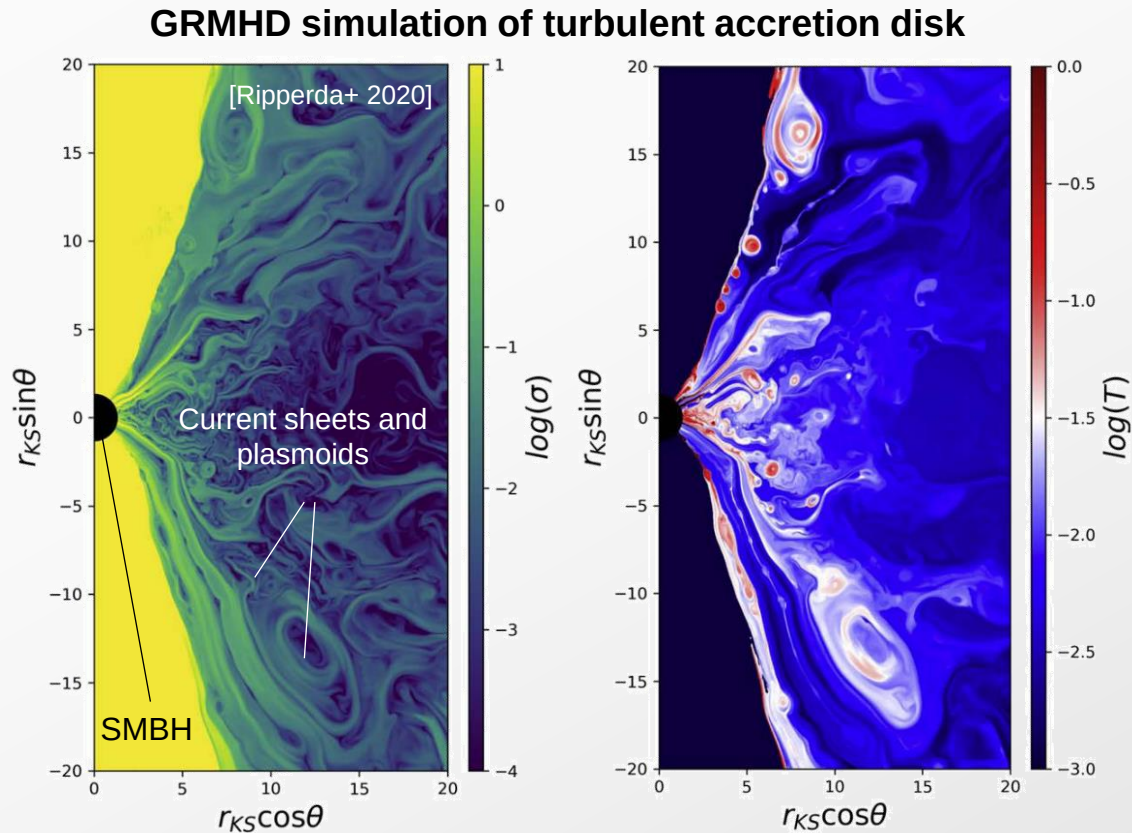
Q: can Spitzer resistivity support these reconnected fields?

$$\eta \propto T_e^{-3/2}$$

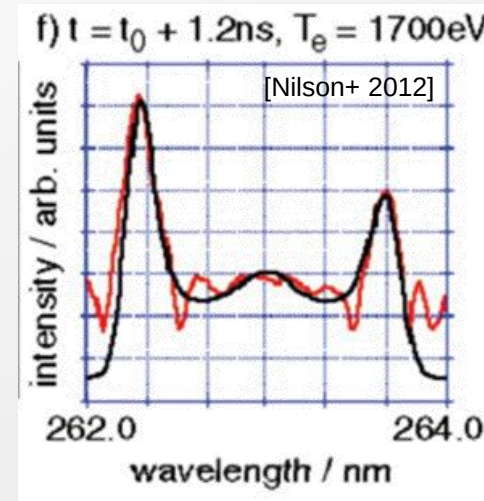
We need to measure electron temperature

The issue of electron heating (and energy partition in general) is interesting in a number of physical systems

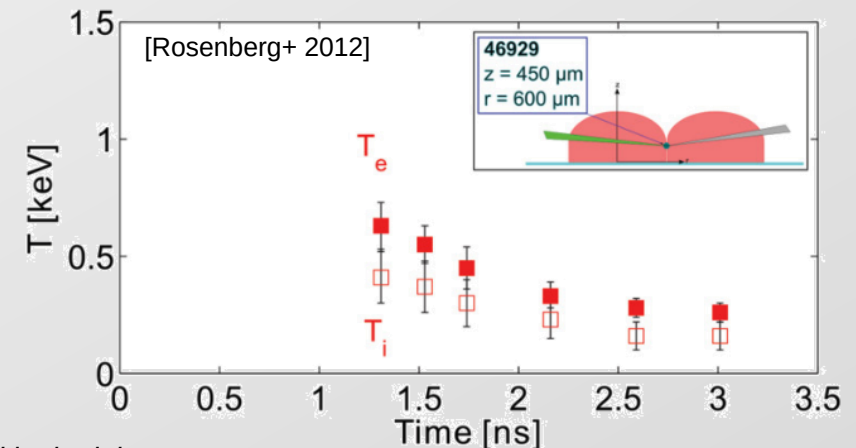
In astrophysics



In the lab

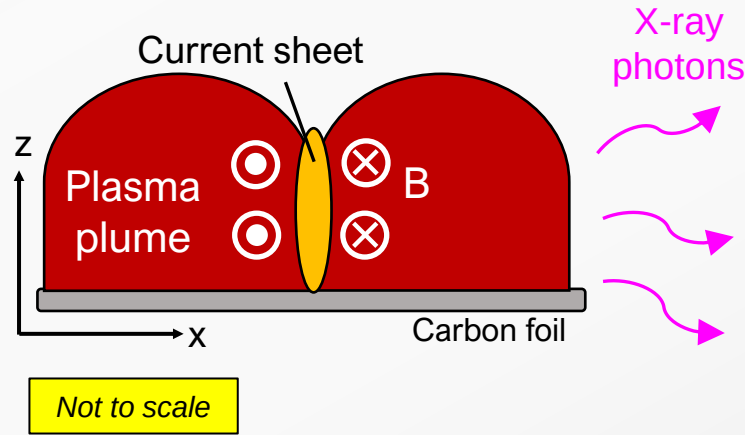


- **Electron heating is laser-driven reconnection is an open question;** prior results disagree
- **We used X-ray imaging to infer electron temperature,** rather than Thomson scattering as a complementary approach

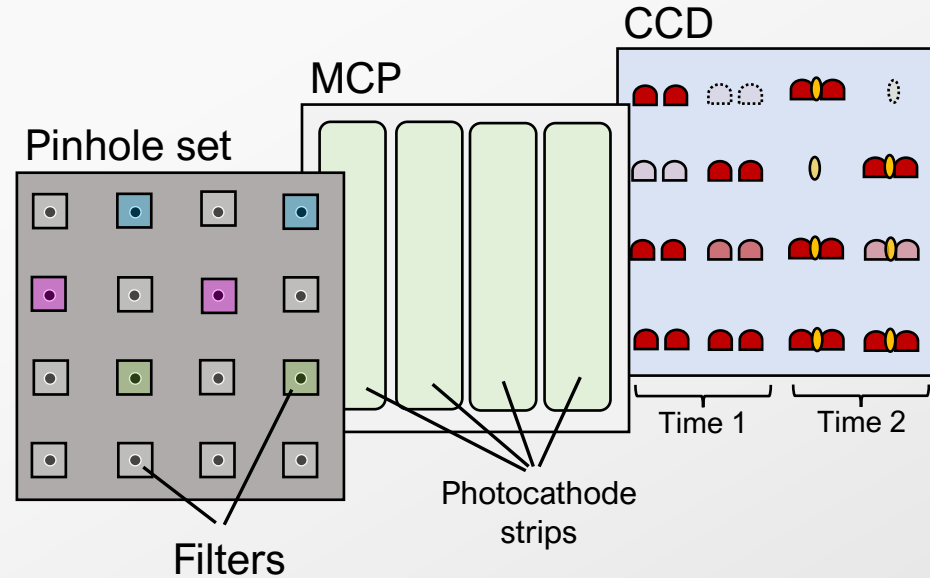


We used a gated X-ray detector (GXD) to obtain 16 filtered self-emission images on each experiment

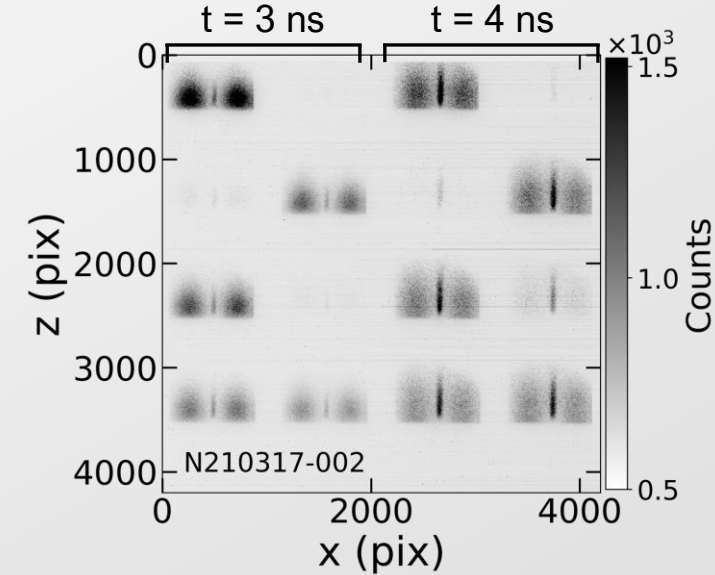
Diagnostic line of sight



GXD configuration



X-ray self-emission images



Spectral emissivity dominated by bremsstrahlung continuum

$$j_\nu \propto \frac{Zn_e^2}{\sqrt{T_e}} e^{-\frac{h\nu}{k_B T_e}}$$

- Diverse filter selection
- **Multiple redundant images analysed** (statistics, uncertainty, robustness)
- Statistical approach can distinguish diagnostic artefacts from plasma structures

Self-emission intensity ratio through different filters is a function of line-averaged electron temperature

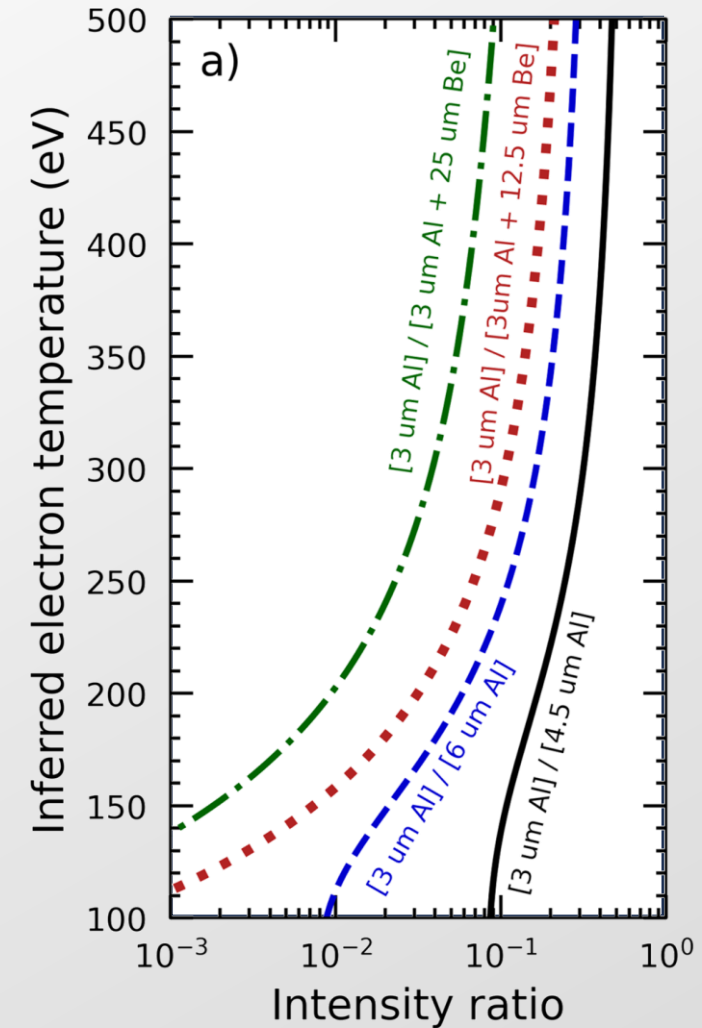
$$\frac{I_1}{I_2}(x, z) = f(\langle T_e(x, z) \rangle_Y)$$

Taking intensity ratios between images with different filters allows to infer electron temperature

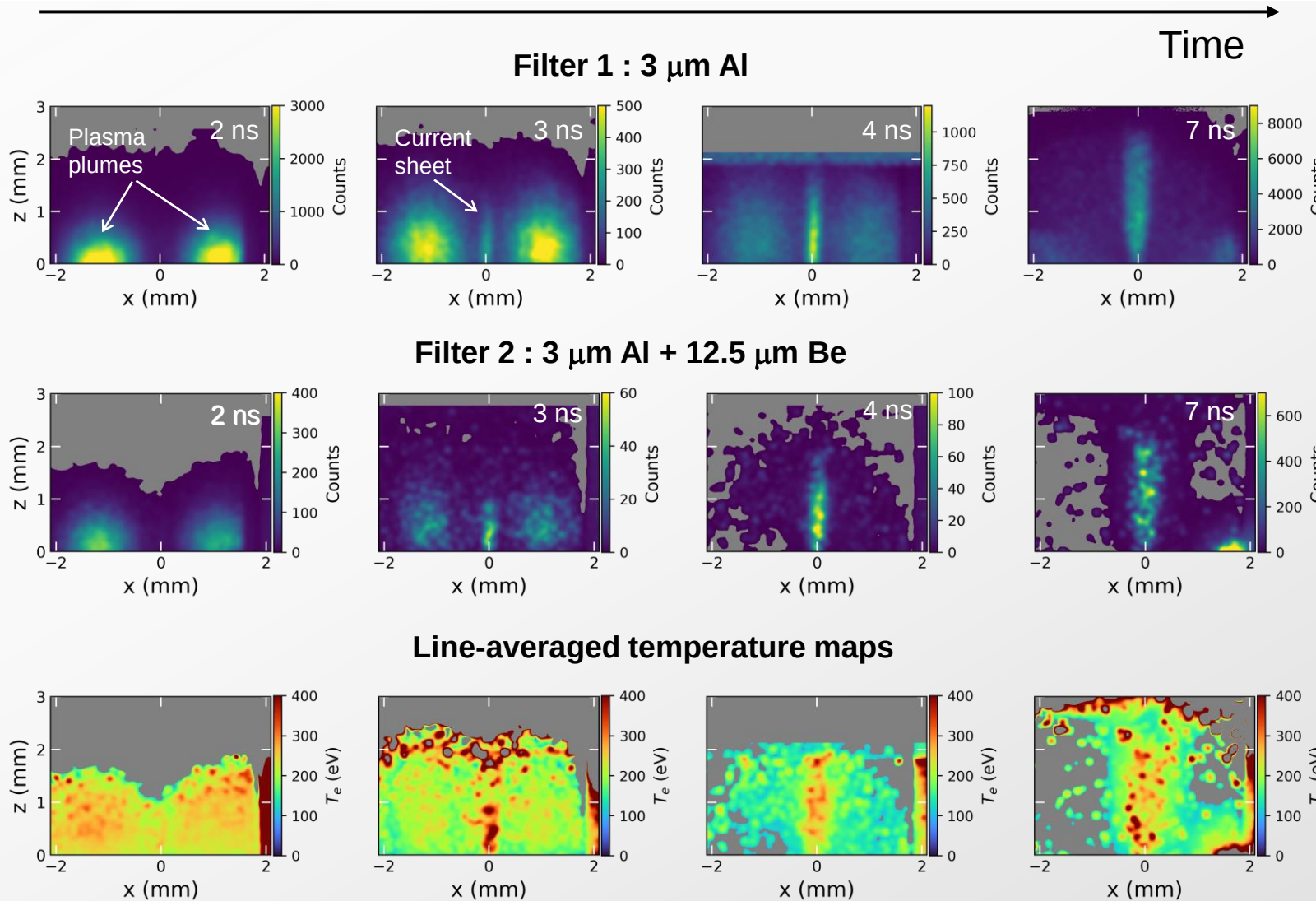
$$\frac{I_1}{I_2}(x, y) = \frac{\int K(\nu)W_1(\nu)j_\nu(x, y)d\nu}{\int K(\nu)W_2(\nu)j_\nu(x, y)d\nu} = f(\langle T_e(x, z) \rangle_Y)$$

$K(\nu)$: GXD response; $W(\nu)$: filter transmission;
 j_ν : plasma emissivity

- Using instrumental response and filter transmission we have calculated a look-up tables to analyse data
- Our code solves for electron temperature at each pixel on the images, building a 2-D map



X-ray images reveal plasma structure, evolution and stability; however finite photon statistics produce artifacts



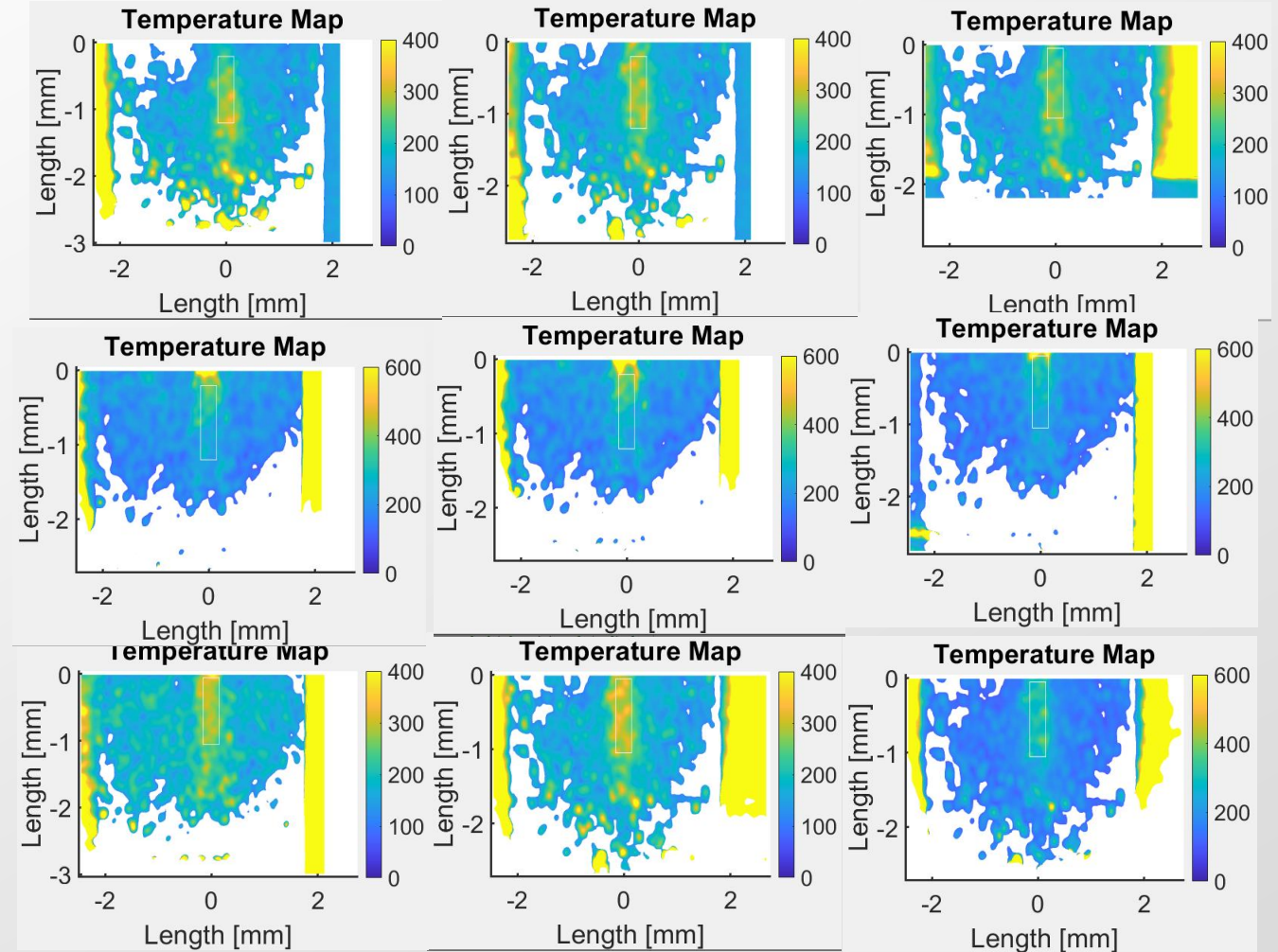
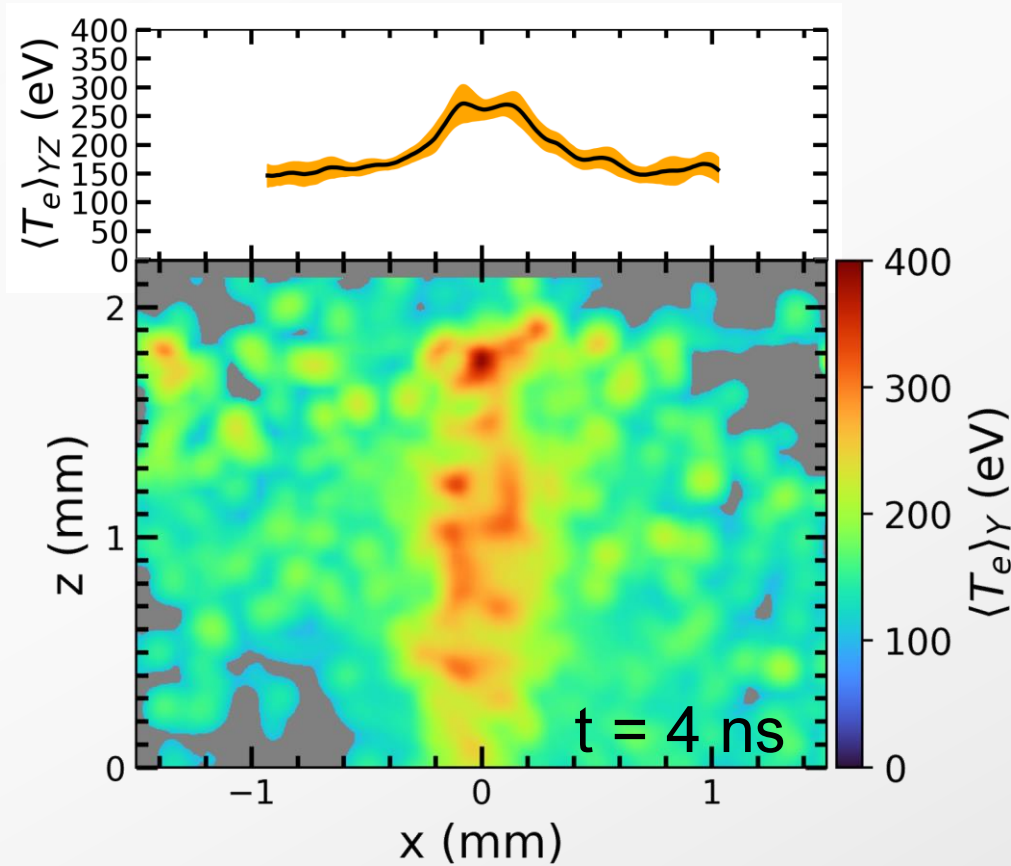
- Current sheet forms at $t \sim 3$ ns
- Inferred inflow velocity $V_{\text{in}} \sim 220$ km/s (consistent with proton radiography estimates)
- Axial expansion at $V_z \sim 300$ km/s ($M_s \sim 1$)

- Enough photon fluence to form images
- **Speckled images due to finite photon statistics effects**

- Gaussian filter for image analysis (FWHM = $115 \mu\text{m} < 200 \mu\text{m}$ spatial resolution)
- **Apparent hot-spots emerge from finite photon statistic effects**

We have calculated several temperature maps for each timing: how do we add all this information together?

All $\langle T_e \rangle_Y$ maps at $t = 4$ ns



Uncertainty dominated by artificial hot-spots. Taking weighted spatial averages allows to construct a statistical assembly

Statistical ensembles

1. For each temperature map

$$\langle T_e \rangle = \frac{1}{N_x N_y} \sum_{i,j} \langle T_e \rangle_Y(x_i, z_j)$$

Spatial average

$$\delta T_e \propto \sqrt{\sum_{i,j} (\langle T_e \rangle_Y(x_i, z_j) - \langle T_e \rangle)^2}$$

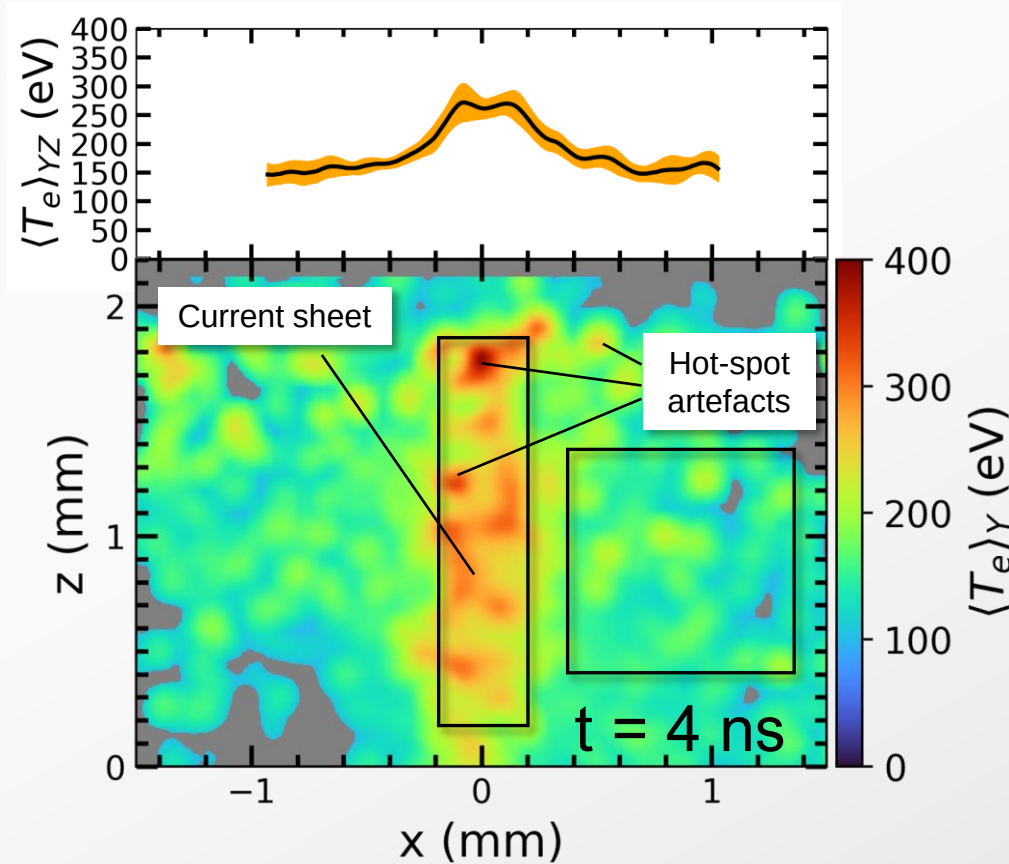
Spatial std. deviation

2. Average temperature at a given time calculated from all temperature maps

$$\bar{T}_e(t) = \frac{\sum_j \omega_j(t) \langle T_e \rangle_j(t)}{\sum_j \omega_j(t)}; \quad \omega_j = \frac{1}{\delta T_{e,j}}$$

Ensemble weighted
average

Weights



Analysis on the plume and reconnection layer at different timings indicate thermal decoupling at 4 ns

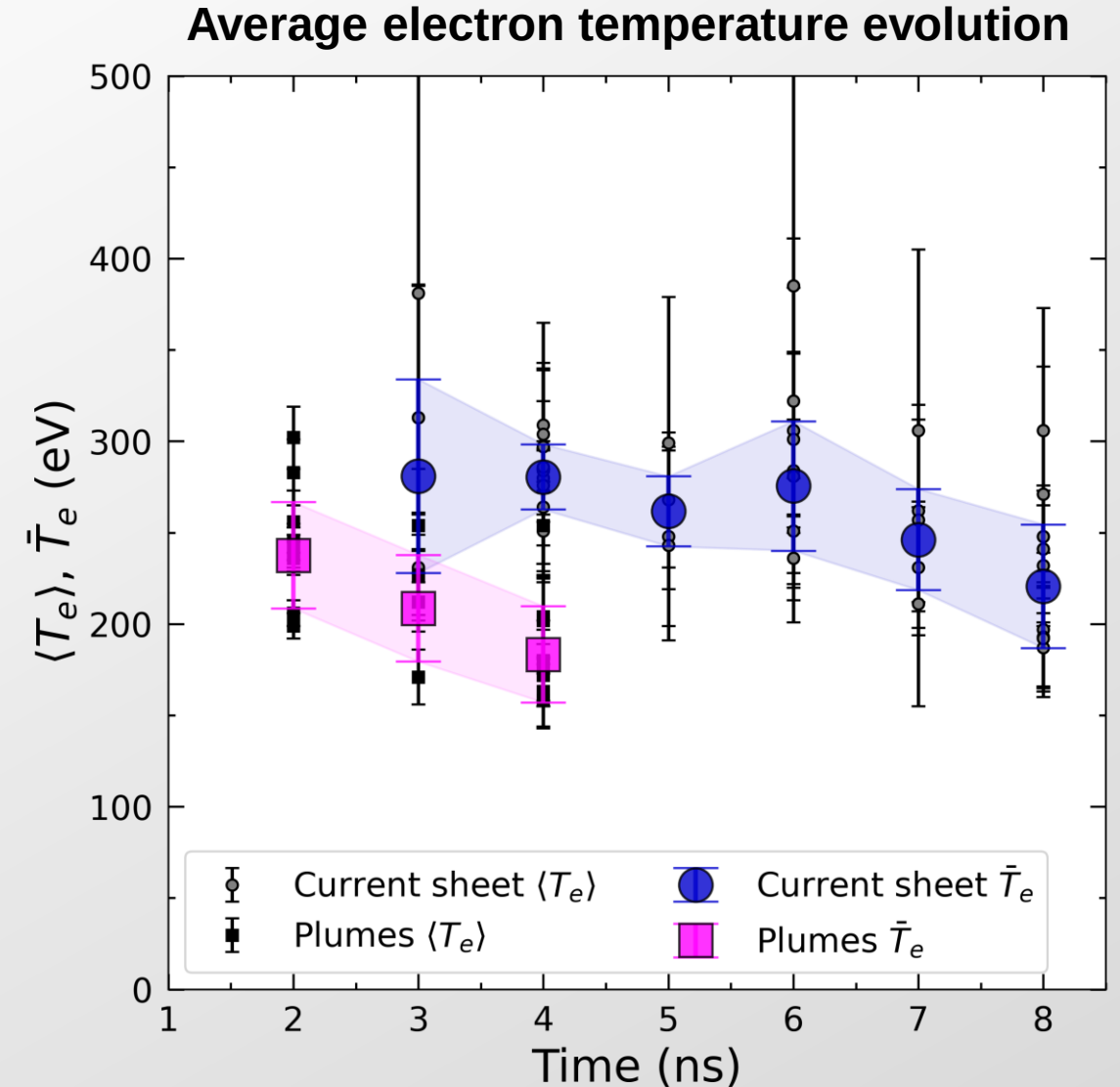
Grey markers: spatial average (individual images)

Coloured markers: ensemble statistical average

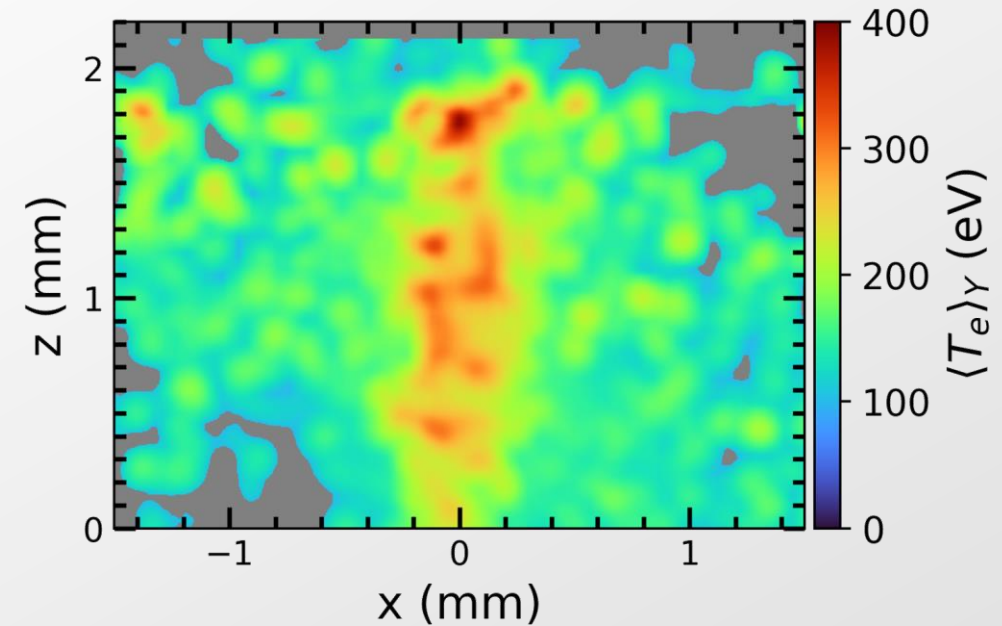
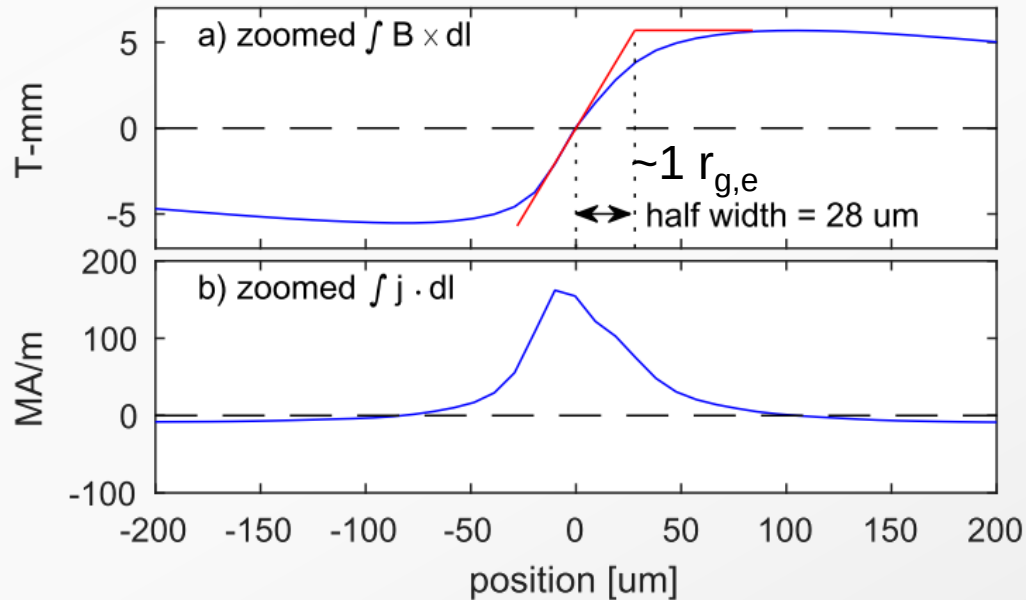
- Averaging over multiple maps drastically decreases uncertainty
- Plasma exhibits slow cooling overall
- At $t = 4$ ns the temperature difference is larger than uncertainty
- **The current sheet is 50% hotter than inflows, indicating significant electron heating**

Considering $V_{in} \sim 220$ km/s, and assuming $n_e = 1e19$ cm $^{-3}$

- *High-beta, flow-dominated regime:* $\beta_{th} \sim 10$, $\beta_{dyn} \sim 50$
- *Free-streaming ions:* $\lambda_{fp,ij}^{mfp} \sim 1.5$ mm > layer width ~ 30 μ m



Going back to the reconnected fields problem: at these temperature, can Spitzer support E and J?



Reconnected line-integrated electric field

$$\langle \int E_z d\ell \rangle = \Delta\Psi/\Delta t \sim 1300 \text{ V}$$

Reconnected line-integrated current density

$$\int j_z d\ell \sim 160 \text{ MA/m}$$

Spitzer resistivity

$$T_e \sim 280 \text{ eV} \rightarrow \eta = 10^{-4} \Omega \text{ cm}$$

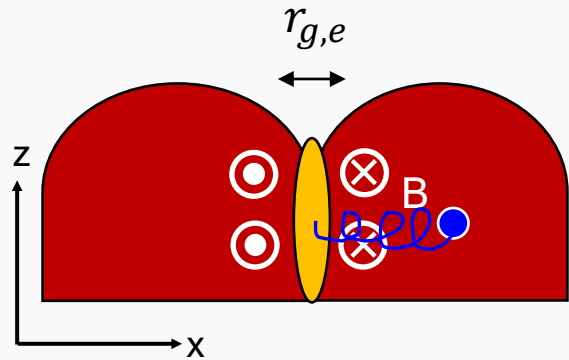
Spitzer reconnected electric field

$$E_{\text{rec}} = \eta \int j_z d\ell \sim 80 \text{ V} \ll E_z \sim 1000 \text{ V}$$

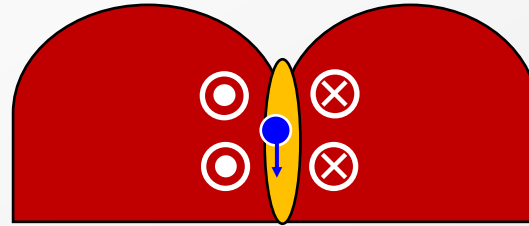
Observed reconnection rate cannot be supported by resistive dissipation

The narrow current sheet and high reconnection rate suggests that kinetic effects can become important

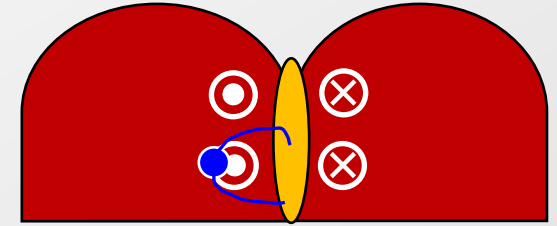
- Current sheet is extremely narrow: $\delta \sim r_{g,e} \rightarrow$ kinetic effects



1. Magnetized electron enters the current sheet



2. Electron becomes demagnetized and is accelerated by reconnected electric field



3. Electron can come back to the inflow and gyrate back. Meandering motion provides effective resistivity.

- Hesse+ 1999 : momentum transport can support a large reconnected electric field

$$E_{\text{rec}} \approx \frac{\sqrt{2m_e T_e}}{e} \frac{\partial v_e}{\partial x} \sim 1000 \text{ V/mm}$$

In the same conditions, this kinetic effect can explain the higher flux annihilation rates we observe

Kinetic effects can also account for the observed electron heating

0-D Electron heating equation

$$\frac{3}{2} n_e k_B \frac{dT_e}{dt} = Q_{\text{comp}} + Q_{\text{rec}} + Q_{\text{col}}$$

- Adiabatic cooling

$$Q_{\text{comp}} = \frac{dn_e}{dt} T_e$$

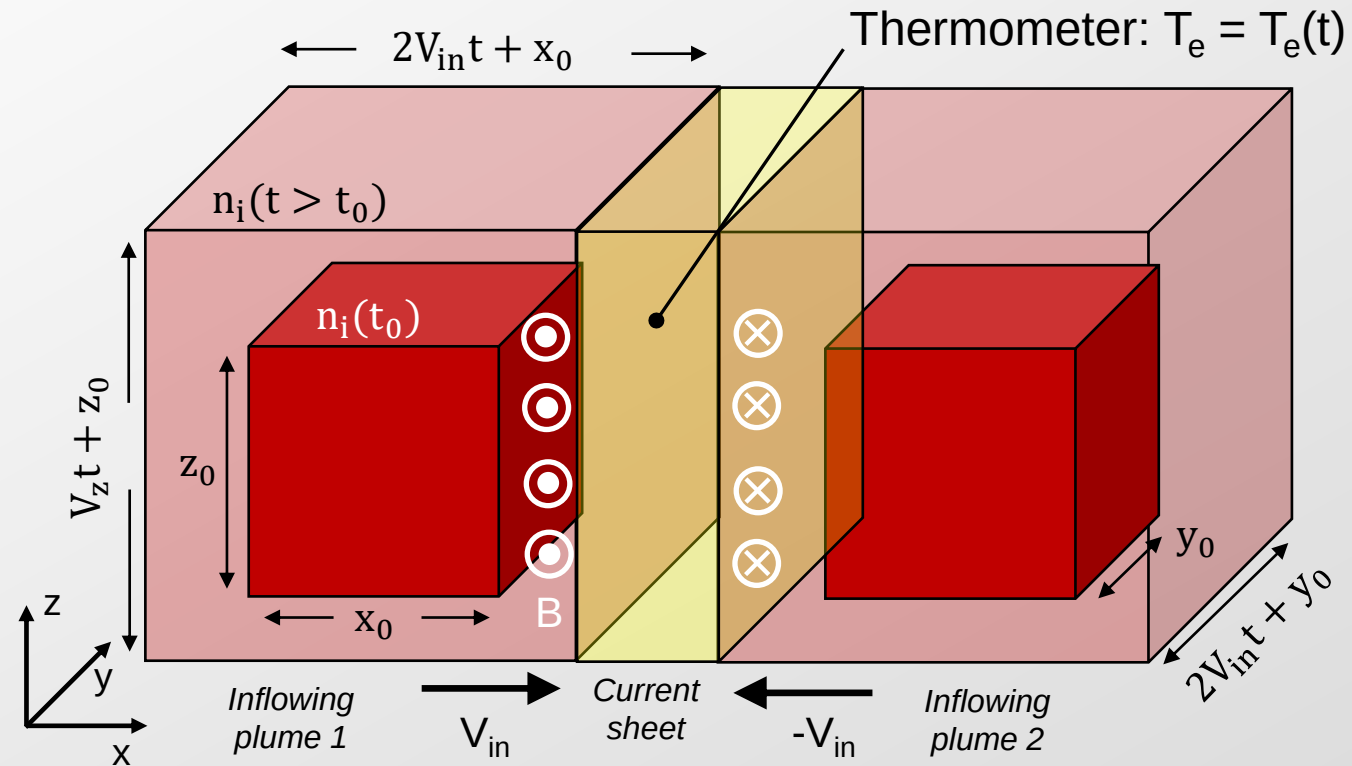
- Magnetic reconnection heating

$$Q_{\text{rec}} = \mathbf{E} \cdot \mathbf{j}$$

- Electron-ion collisional heating [Ross+ 2012; Ryutov+ 2011]

$$Q_{\text{col}} = 4m_e v_{ei} n_e V_{\text{in}}^2$$

Ballistic expanding box model



Several mechanisms can play a role in the current sheet heating. For $n_e > 7e18$ pcc, collisional heating is dominant

0-D Electron heating equation

$$\frac{3}{2} n_e k_B \frac{dT_e}{dt} = Q_{\text{comp}} + Q_{\text{rec}} + Q_{\text{col}}$$

- Adiabatic cooling

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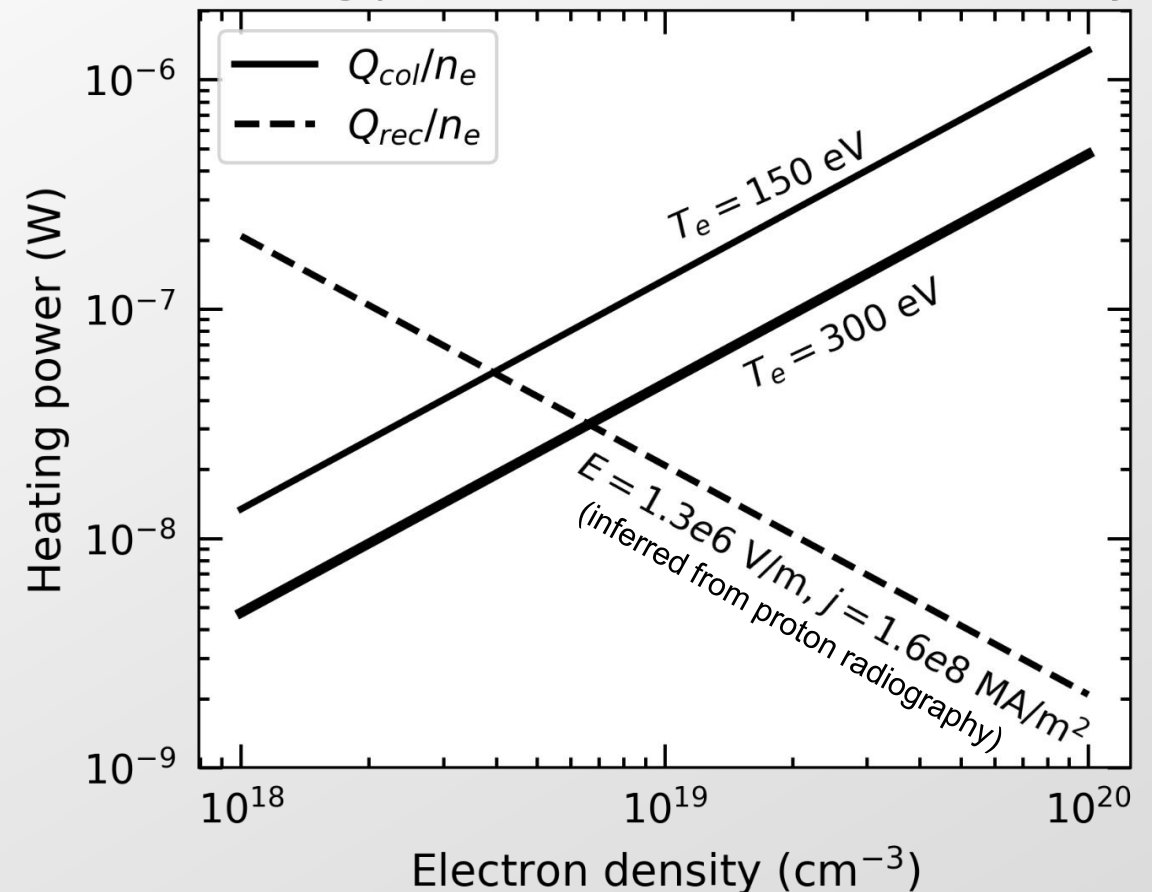
- Magnetic reconnection heating

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a) Collisional drag and magnetic reconnection heating power as function of inflow density



Electron-ion collisional heating is sufficient to heat up the reconnection layer in a broad density range

0-D Electron heating equation (reduced)

$$\frac{3}{2} n_e k_B \frac{dT_e}{dt} = Q_{\text{comp}} + \cancel{Q_{\text{rec}}} + Q_{\text{col}}$$

has the solution found by [Ross+ 2012]

$$T_e(t)^{5/2} \propto \int_{t_0}^t V_{\text{in}}^2(t') n_i^{-2/3}(t') dt' + T_e(t_0)^{5/2},$$

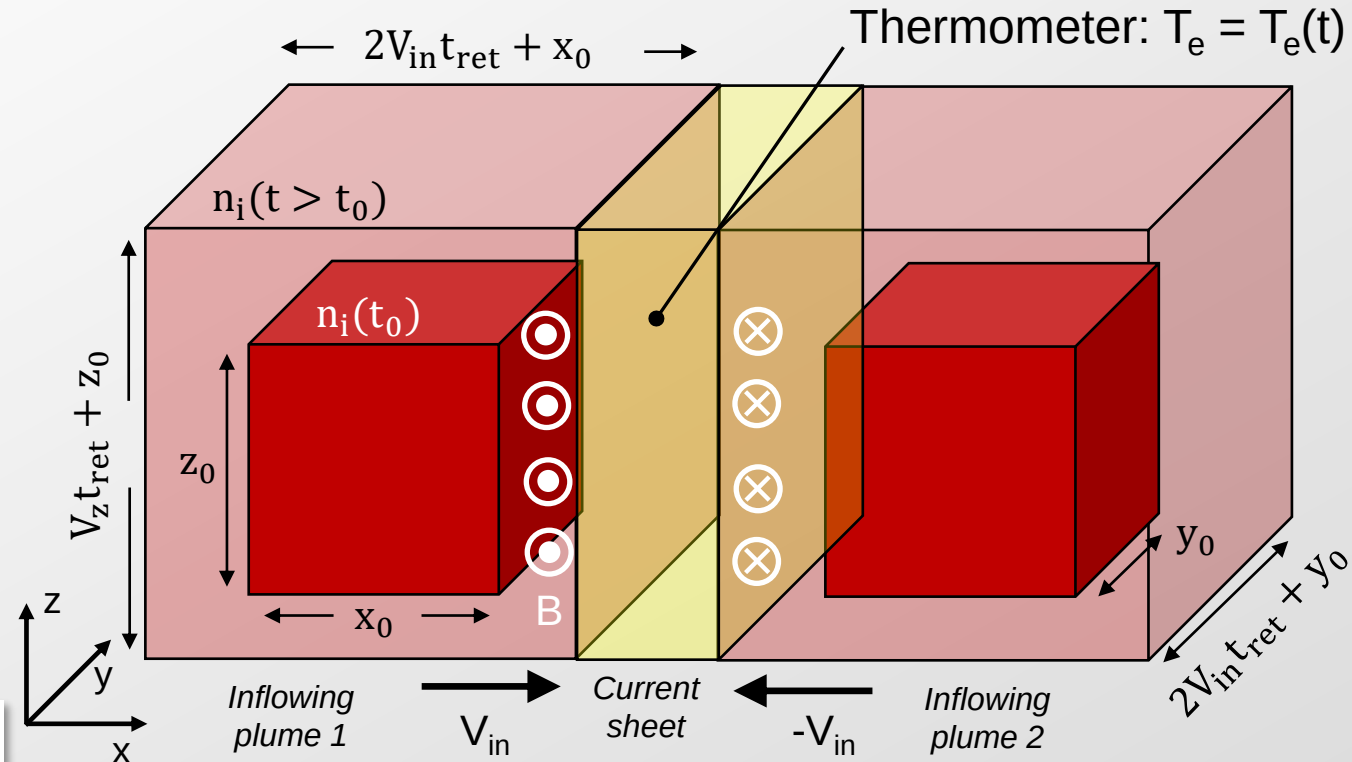
Which can be computed using the first-order ballistic expanding box model parameterising the inflow density

$$n_i(t) = n_i(t_0) \left(\frac{x_0}{V_{\text{in}} t_{\text{ret}} + x_0} \right) \left(\frac{y_0}{V_{\text{in}} t_{\text{ret}} + y_0} \right) \left(\frac{z_0}{V_z t_{\text{ret}} + z_0} \right)$$

with $t_{\text{ret}} = t - t_0$

which has only the initial plume $n_i(t_0) = n_e(t_0)/Z$ as a free integration constant.

Ballistic expanding box model



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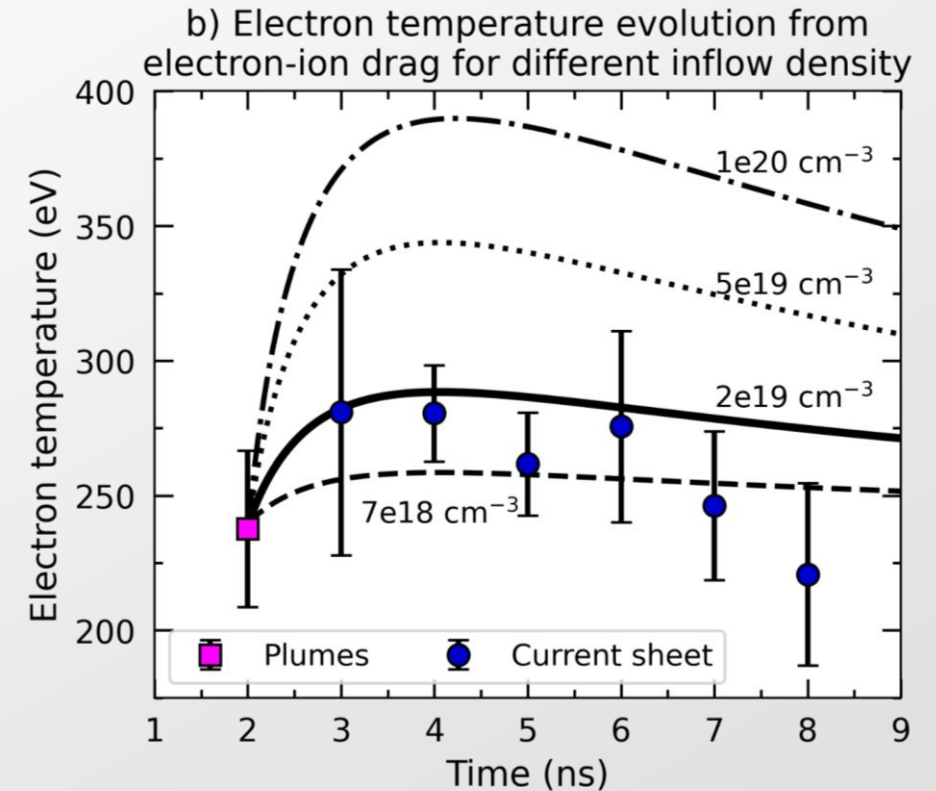
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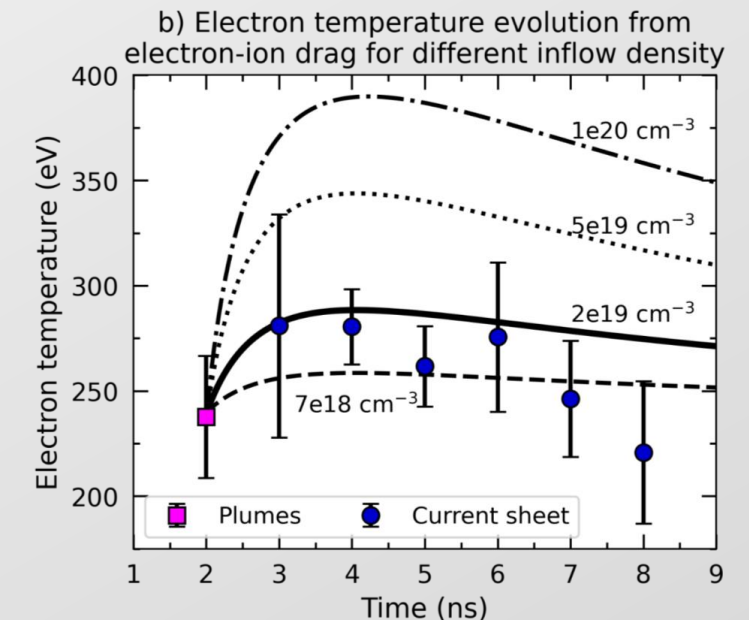
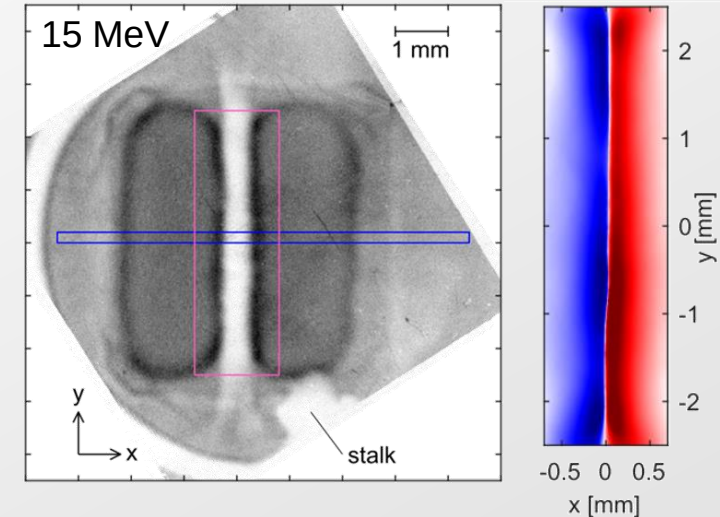
In the high-density regime ($>2e19 \text{ cm}^{-3}$) collisional drag is enough to explain the fast heating

Summary

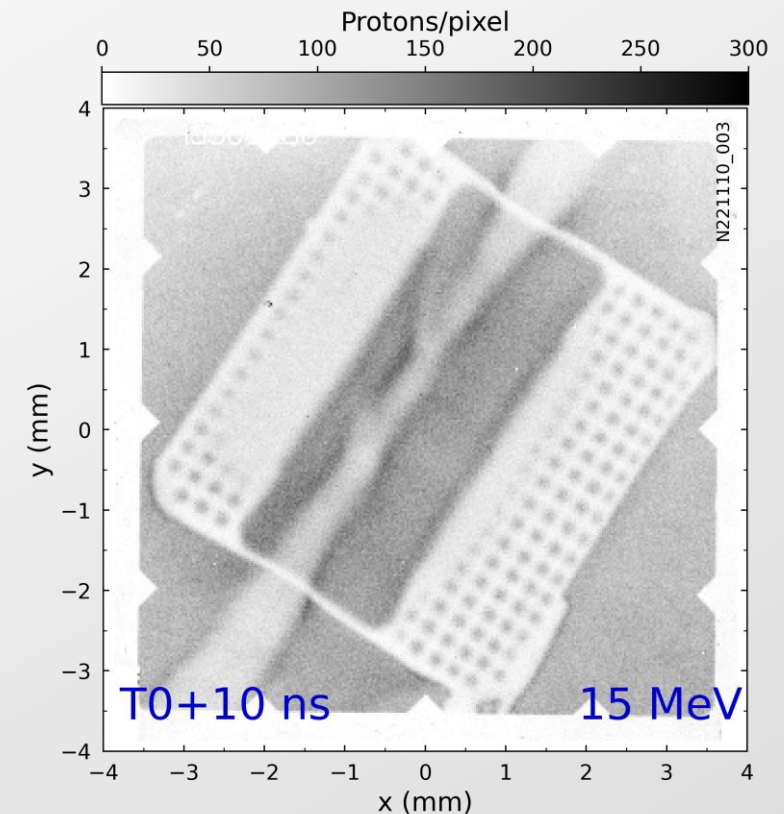
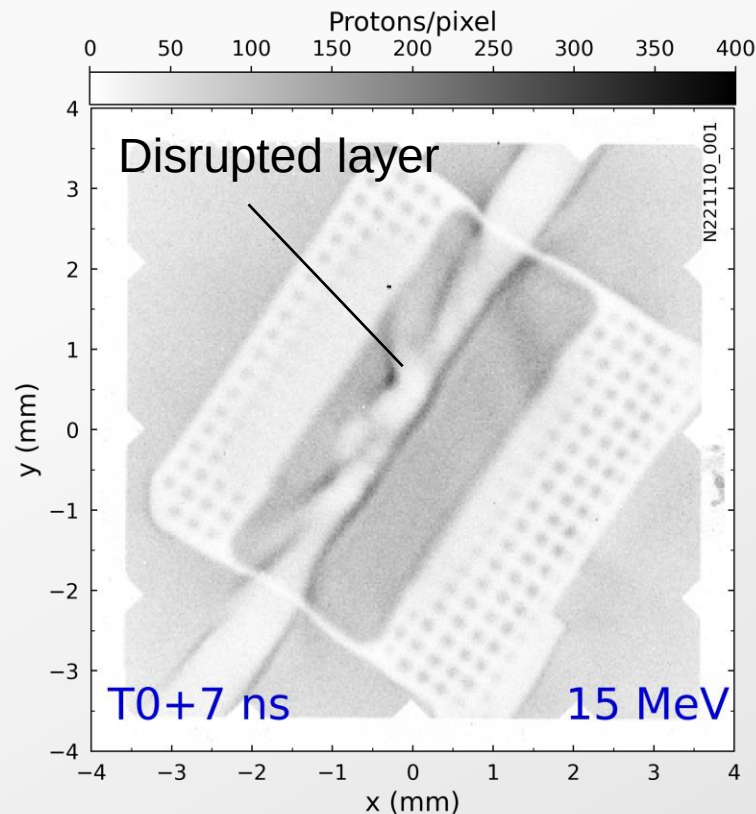
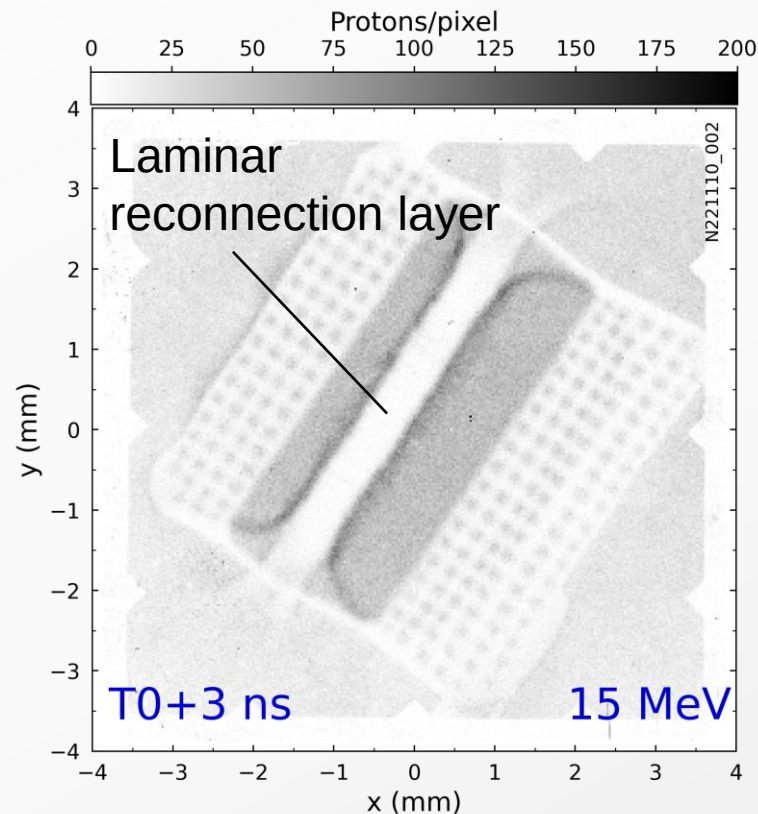
- Magnetic reconnection is a fascinating process ubiquitous in laboratory, space, and cosmic plasmas
- Dedicated high-energy-density experiments can probe properties of reconnecting systems

My take-home message:

- **Our magnetic reconnection experiments seem to be accessing interesting kinetic effects**
 - In the kinetic regime, the reconnection rate can be explained by considering extended effects. Electron-scale physics can speed up the reconnection rate.
 - Competing cooling/heating mechanisms interplay with magnetic reconnection. Electron-ion binary Coulomb collisions can significantly heat the electrons, increasing the reconnection rate.



Future work: plasmoids? Filaments?



- At later times, the current sheet breaks!
- **Nonlinear coupling:** Interplay between **tearing (plasmoid)** and **filamentation (Weibel)** instabilities?
- **Energy partition:** Net effect in particle heating? Heat transport?

Ask me next year! :)

Summary

- Magnetic reconnection is a fascinating process ubiquitous in laboratory, space, and cosmic plasmas
- Dedicated high-energy-density experiments can probe properties of reconnecting systems

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